

IN THIS PART ...

Explore how the body's systems let it move, think, feel, and thrive on a day-to-day basis. Build up nerves; then branch out from the central to the peripheral and autonomic systems.

Get acquainted with mission control — the human brain — and sense the world around the body through the eyes and the ears.

Meet the ringmasters of the endocrine system, including the hypothalamus and the pituitary gland, and balance the body with homeostasis.

Chapter 9

Feeling Jumpy: The Nervous System

IN THIS CHAPTER

- » Breaking down the structure of nerves
 - » Centering with the central nervous system
 - » Branching out with the peripheral nervous system
 - » Taking a hands-off approach to the autonomic nervous system
 - » Examining the senses
-

Throughout this book, you look at the human body from head to toe, exploring how it collects and distributes the molecules it needs to grow and thrive, how it reproduces itself, and even how it gets rid of life's nastier byproducts. In this chapter, however, you look at the living computer that choreographs the whole show, the one system that contributes the most to making us who we are as humans: the nervous system.

In this chapter, you get a feel for how the nervous system is put together. You practice identifying the parts and functions of nerves and the brain itself as well as the structure and activities of the central, peripheral, and autonomic systems. In addition, we touch on the sensory organs that bring information into the human body.

Our Motherboard

The nervous system is the communications network that goes into nearly every part of the body, innervating your muscles, pricking your pain sensors, and letting you reach beyond yourself into the larger world. More than 80 major nerves make up this intricate network, with billions of *neurons* (individual nervous cells). It's through this complex network that you respond both to external and internal stimuli, demonstrating a characteristic called

irritability (the capacity to respond to stimuli, not the tendency to yell at annoying people).

Put another way, there are three functions of the human nervous system as a whole:

- » **Sensation:** The ability to generate nerve impulses in response to changes in the external and internal environments (this also can be referred to as *perception*)
- » **Coordination:** The ability to receive, sort, and direct those signals to channels for response (this also can be referred to as *integration*)
- » **Reaction:** The ability to translate orders to the *effectors* (muscles and glands), which carry out the body's response to the stimulus

The nervous system has two divisions — the *central* and *peripheral nervous systems*. The central nervous system (CNS) includes the brain and spinal cord and is the body's integrator. Everything else falls into the peripheral nervous system. This is split into *sensory* and *motor divisions*. The sensory, or *afferent*, system responds to stimuli and communicates it to the CNS. If a response is warranted, the message is sent to the motor, or *efferent*, system, leading to the body's response to the stimulus. The motor system is further divided into the *somatic nervous system* which carries out voluntary commands and the *autonomic nervous system* which carries out all the automatic functions that we don't have to think about.

Before trying to study the system as a whole, it's best to break it down into building blocks first: neurons, nerves, impulses, and synapses.

Building from Basics: Neurons, Nerves, and Glial Cells

The nervous system is comprised mostly of neural tissue along with connective tissues for structure and to house blood and lymphatic vessels. There are two types cells found in neural tissue: *neurons* and *glial cells* (*neuroglia*). *Nerves* are not cells, they are an organizational structure that bundles together extensions of the neurons called axons.

Neurons

The nervous cell responsible for the communication at the core of the nervous system's functions is the neuron. Its properties include that marvelous *irritability* that we speak of in the chapter introduction as well as *conductivity*, otherwise known as the ability to transmit a *nerve impulse*.

The central part of a neuron is the *cell body*, or *soma*, that contains a large nucleus with one or more nucleoli, the organelles common to all other cells, and *Nissl bodies*. (See [Chapter 3](#) for an overview of a cell's primary parts.) The Nissl bodies are also found in the dendrites and are specialized pieces of rough ER thought to make proteins for the cell itself. The ER found in the cell body manufactures *neurotransmitters* that the neuron uses to communicate with other cells. Extending from the cell body are numerous cytoplasmic projections, containing specialized fibrils, or *neurofibrillae*, forming *dendrites* and *axons*. Dendrites conduct impulses to the cell body while *axons* conduct impulses away from it (see [Figure 9-1](#)). While a neuron can have numerous dendrites, it has only one axon; however, each axon can have many branches called *axon collaterals*, enabling communication with many target cells. The point of attachment on the cell body is called the *axon hillock*. The opposite end of the axon enlarges into a *synaptic knob* or *bouton*.

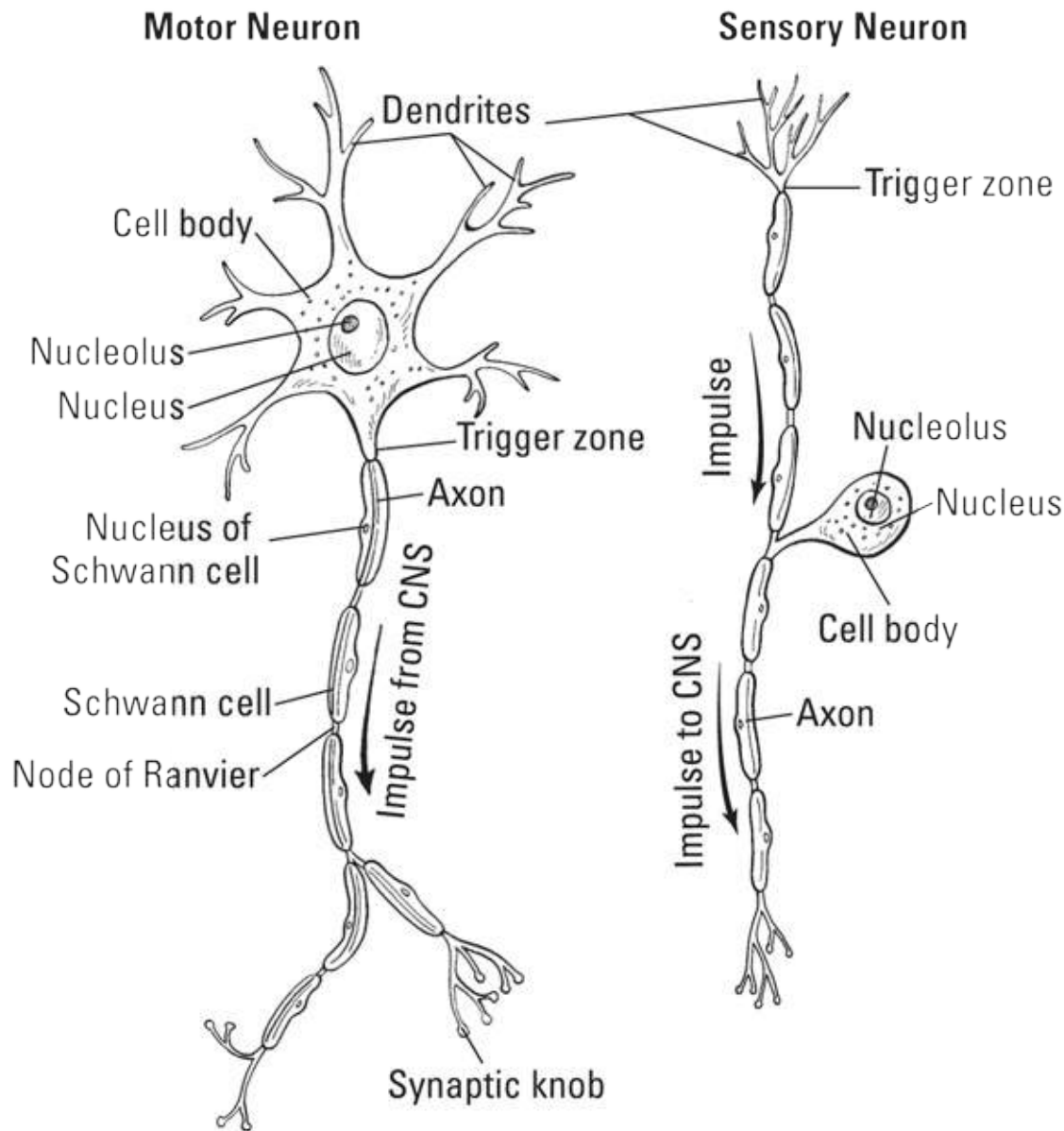


Illustration by Kathryn Born, MA

FIGURE 9-1: The motor neuron on the left and the sensory neuron on the right show the cell structures and the paths of impulses.



REMEMBER There are four types of neurons, as follows:

- » **Multipolar:** Multiple dendrites separated from the axon by the cell body. Most motor, or efferent, neurons are of this type, connecting the CNS to its effectors — the structures that will carry out the effect. (Get it?)

- » **Unipolar:** Both the axon and dendrites come from the same cytoplasmic extension, meaning the dendrites connect directly to the axon rather than through the cell body. Most sensory, or afferent, neurons are unipolar carrying information about stimuli to the CNS.
- » **Bipolar:** A single dendrite and single axon (though they both have branches) meet the cell body on opposite sides. These are found almost exclusively in the CNS, the only exception being specialized sensory neurons from the eyes, nose, and ears.
- » **Anaxonic:** Axon cannot be distinguished from dendrites. These neurons are found in the brain, but their function is not well understood. They are thought to regulate motor neuron networks and may play a role in learning.



TIP

Here are a couple of handy memory devices: Afferent connections arrive, and efferent connections exit. Dendrites deliver impulses while axons send them away.

Nerves



REMEMBER

Whereas neurons are the basic unit of the nervous system, *nerves* are the cable-like bundles of axons that weave together the peripheral nervous system. There are three types of nerves:

- » **Afferent nerves** are composed of sensory nerve fibers (axons) grouped together to carry impulses from receptors to the central nervous system.
- » **Efferent nerves** are composed of motor nerve fibers carrying impulses from the central nervous system to effector organs, such as muscles or glands.
- » **Mixed nerves** are composed of both afferent and efferent nerve fibers.

The diameter of individual axons (nerve fibers) tends to be microscopically small — many are no more than a micron, or one-millionth of a meter. But

these same axons extend to lengths of 1 millimeter and up. The longest axons in the human body run from the base of the spine to the big toe of each foot, meaning that these single-cell fibers may be 1 meter (about three feet) or more in length.

There are 12 pairs of cranial nerves that relay information from the head and neck to the brain. There are 31 pairs of spinal nerves that do the same for the rest of the body.

From the inside out, nerves are composed of the following:

- » **Axon:** The impulse-conducting process of a neuron, also known as a fiber.
- » **Myelin sheath:** An insulating envelope that protects the axon and speeds up impulse transmission. (This process is discussed in detail in the [“Impulses”](#) section later in this chapter.)
- » **Neurolemma (or neurilemma):** A thin membrane present in many peripheral nerves that surrounds the myelin sheath and its enclosed axon.
- » **Endoneurium:** Loose, or *areolar*, connective tissue surrounding the length of individual fibers.
- » **Fasciculi:** Bundles of fibers within a nerve.
- » **Perineurium:** The same kind of connective tissue as endoneurium; bundles fascicles together.
- » **Epineurium:** The same kind of connective tissue as endoneurium and perineurium; surrounds all the bundles, creating the nerve.

Nerves organize the transport of signals both to and from the brain. Think about traveling across the country by car. Which is more efficient: taking an interstate or city streets? Nerves are the interstates of our nervous system's roads. Since nerves contain only the axons of peripheral neurons, a different structure organizes the cell bodies (and their dendrites). This structure, called a *ganglion*, bundles the cell bodies of related neurons (both afferent and efferent). These are found as the cranial and spinal nerves enter the CNS. The ganglia connect with each other forming a *plexus* that controls related body functions. For example, the solar plexus controls contractions throughout the digestive tract.

Glial cells

Much has been learned about glial cells in the past 30 years. At first, scientists thought that glial cells merely held neurons in their proper position, hence their name. (*Glia* is Greek for glue.) Until recently, they had been largely ignored by researchers, assuming they provided only structural support. However, we have learned that these “other nervous cells” have some pretty important functions. [Table 9-1](#) describes the types of glial cells and their functions.

Table 9-1 Cranial Nerves

<i>Glial Cell</i>	<i>Description</i>	<i>Function</i>
Astrocyte	Star-shaped, found between neurons and blood vessels in CNS	Form blood brain barrier, takes up excess neurotransmitters and ions, damage repair, create new neuron connections
Ependyma	Often ciliated, forms simple cuboidal epithelium in CNS	Creates a filter controlling which molecules can move in and out of the CSF and the nervous tissue
Microglia	Small cell with tiny branches in CNS	Phagocytosis of debris, waste, and pathogens
Oligodendrocyte	Rounded cell with extensions that wrap around axons in the CNS	Form myelin sheaths, produce neurotrophic factors (growth and development of neurons)
Schwann cell	Small cell with extensions that wrap around axons in the PNS	Form myelin sheaths
Satellite cell	Flat cell surrounding cell bodies in ganglia of PNS	Protection, regulate nutrients

As you can see from [Table 9-1](#), glial cells do far more than just being the glue that holds neurons together! Now let’s see if you can answer the following questions about glial cells and the other cells and structures of the nervous system:

1-5 Fill in the blanks.

The communicating cells of the nervous system, or **1.**_____ have two essential properties. First is **2.**_____, the tendency to respond to a stimulus. Second is the ability to generate electricity, called **3.**_____. This flow of electricity is known as **4.**_____ but cannot be used to communicate with other neurons. For this, **5.**_____ are used.

6 What are the three primary functions of the nervous system?

- a. Motion, circulation, conceptual thought
- b. Action, conduction, filtration
- c. Perception, integration, action
- d. Circulation, coordination, response

7 Which structure is found at the end of an axon?

- a. Soma
- b. Dendrite
- c. Nissl body
- d. Ganglion
- e. Synaptic knob

8 Which part of a neuron is responsible for conducting electrical signals toward the cell body?

- a. axon
- b. soma
- c. bouton
- d. dendrite
- e. hillock

9 Which membrane surrounds an axon (nerve fiber)?

- a. Sarcolemma
- b. Neurolemma
- c. Perineurium
- d. Fasciculus
- e. Epineurium

10-15 Match the term to its description.

- a. Cells that form and preserve myelin sheaths in PNS
- b. Cells that form and preserve myelin sheaths in CNS
- c. Cell that provide nutrients for bundled cell bodies
- d. Cells that are phagocytic
- e. Cells that form linings in the CNS
- f. Cells that contribute to the repair process of the central nervous system

10. _____ Astrocytes

11. _____ Microglia

12. _____ Oligodendrocytes

13. _____ Schwann cell

14. _____ Satellite cell

15. _____ Ependyma

16 How many dendrites and axons are there on a multipolar neuron?

- a. Many dendrites and one axon
- b. Three dendrites and three axons
- c. Four or more dendrites and two axons
- d. A single dendrite and a single axon
- e. Numerous dendrites and numerous axons

17 A synapse between neurons is best described as the transmission of

- a. a continuous impulse.
- b. an impulse through chemical and physical changes.
- c. an impulse through a physical change.
- d. an impulse through a chemical change.

18 The connective tissue that surrounds a bundle of fibers in a nerve is known

as the

- a. Neurolemma
- b. Perineurium
- c. Endoneurium
- d. Fasciculus
- e. Epineurium

19 True or False: Nervous tissue is essentially neurons connected end-to-end with very little matrix.

20 Which statement best describes how neurons communicate?

- a. Axons generate impulse which is passed to dendrites through the cell body.
- b. Axons receive a stimulus that is passed to the cell body, triggering the dendrite to release a chemical.
- c. Cell bodies respond to a stimulus by generating an electrical signal to pass on to dendrites and axons, where the former will pass it on chemically and the latter will do so electrically.
- d. In response to stimulus, the dendrites generate an electrical signal that is passed to the cell body. The axon will then spread the impulse down its length releasing a chemical at the end.

Feeling Impulsive?

Neurons communicate using both electric and chemical signals. The electrical signal comes from the movement of ions — much like the flow of electrons we know as electricity. The chemical signal comes in the form of *neurotransmitters*. The entire communication process occurs in three phases: stimulus of the dendrites, impulse through the axon, and synaptic transmission from the synaptic knob.

Along the neuron

Before diving in to the communication process, you must first understand that

a neuron is a *polarized* cell. Compared to its surroundings, the inside of the cell is more negatively charged — to the tune of -70mV . This is due to the high concentration of *anions* (negatively charged molecules) held there, which the neuron maintains when at rest, hence the term *resting potential*. Additionally, the axon has a high concentration of potassium ions (K^+) inside and a high concentration of sodium ions (Na^+) outside.



TIP

Don't let the positive charge of potassium ions throw you off. There are still numerous organic anions that counteract the K^+ inside of the cell, creating the resting potential of -70mV .

Stimulus

The stimulus of a neuron generally happens in one of two ways. Most commonly, it is done by a neurotransmitter, a chemical messenger released from another neuron. The neurotransmitter binds to receptors on the dendrites, which triggers ion channels to open. The other option occurs only in sensory receptor neurons. Here, the designated stimulus causes the ion channels to open — for instance, heat in a hot thermoreceptor (but not cold!) or touch in a mechanoreceptor. Regardless of how the channels are triggered to open, positive ions rush into the cell (it's -70mV in the cell, after all). This causes the *depolarization* of the cell body as it becomes more positive (but overall still negatively charged). This initial depolarization, however, does not mean the neuron will proceed in communicating the signal. For that to occur, enough positive ions must come in to depolarize the cell body, where it meets the axon, to the threshold potential, which is -55mV (on average). If threshold is not reached, the ions are pumped out of the cell body and that's it.



REMEMBER

A single stimulus is generally not enough to warrant the neuron sending a message (just think of how exhausting it would be if we bombarded the brain with random, useless information). A larger (or repeated) stimulus will open more channels, causing more positive ions to rush in, and threshold is more likely to be reached. This causes the *all-or-none response* of a neuron. It either sends a message or it doesn't.

Value of sensation (“this is warm” versus “whoa that’s really hot!”) is imparted by the brain based on how many neurons are sending the same message.

Impulse

The axon hillock is the *trigger zone* for impulse — the trigger being the reaching of threshold potential. This section of the axon has numerous *voltage-gated* sodium channels that pop open when the cell reaches -55mV . This is the beginning of the first *action potential*. A single action potential occurs in the following steps:

1. Sodium gates open.
2. Na^+ rushes in, depolarizing that section of the axon.
3. This continues until enough Na^+ comes in to trigger the voltage gated channels to close, which is $+35\text{mV}$ on average.
4. Reaching 35mV also triggers potassium gates to open and K^+ flows out.
5. The axon segment is now repolarized to the resting potential (-70mV).
6. Too many K^+ leave causing *hyperpolarization* of the cells.
7. The hyperpolarization kick-starts the *sodium-potassium pumps* to move the ions against their concentration gradients to their original locations.

The Na/K pumps are the only step of action potential that requires ATP. Until all the ions have been replaced, the axon segment is in a *refractory period*, meaning it is unable to respond to another stimulus.

An action potential only carries us down one small segment of the axon — nowhere near its entire length. This is where *impulse*, or the sequential generation of new action potentials, comes in. When Na^+ first rushes in, some of them flow down the axon increasing the charge within. Once enough slide over to hit -55mV , a new set of sodium gates open, generating the next action potential. This process repeats all the way to the synaptic knob while the individual segments continue to proceed through the process of action potential.



TIP

Remember in elementary school learning how to sing in the round? You were split into groups and sang “Row, row, row your boat,” but each group started at a different time. This is how impulse works. If you follow the impulse, all you hear is “row, row, row” over and over again until the synaptic knob is reached. But if you only listen to the first group, you hear the entire song — the action potential.

While there is no scientific consensus on how often a neuron can fire, we do have a method to speed up the process: *saltatory conduction*. Many neurons have beads of *myelin* along the length of the axon. These sheets of lipids are created by glial cells and are wrapped around, leaving small sections of axon exposed called the *nodes of Ranvier*. When Na^+ enters the axon they continue to flow down under the insulation of the myelin. A new action potential can only be initiated at the nodes (because any Na^+ gates would be covered up). Thus, a myelinated neuron needs far fewer action potentials to span the length of the axon, saving energy (less ion pumping for recovery) and increasing signal speed up to 100 times.

Across the synapse

Unfortunately, most neurons cannot pass on an electrical signal to another cell, so our communication process is not yet complete. Instead, the electrical signal must be converted into a chemical one; this occurs at the synaptic knob.

When impulse reaches the end of the axon, the final action potential brings in an influx of Na^+ as before. The immediate area depolarizes, but instead of opening sodium gates, calcium gates are opened instead (also triggered by reaching -55mV). Calcium ions rush into the synaptic knob where neurotransmitters are stored in vesicles. The calcium ions shove the vesicles to the cell membrane and exocytosis occurs (see [Chapter 3](#)), releasing the neurotransmitters into the *synapse* (the small space between the cells; they aren’t physically connected). The *pre-synaptic* (sending) neuron has now completed its task and the *post-synaptic* (receiving) cell now repeats the process.



REMEMBER Neurotransmitters can be excitatory or inhibitory. The previous explanation assumes that the stimulating neurotransmitters are excitatory, opening channels for positive ions to enter the dendrites. Some neurotransmitters, however, are inhibitory, either closing channels or opening ones for negative ions to rush in. If a neuron had been repeatedly firing but was then stimulated by enough inhibitory neurotransmitters, it would no longer reach threshold and would stop its communication.

Now let's get your neurons firing to answer some questions.

21-25 Match the term to its description.

- a. A differential of charge from outside of the cell to the inside
- b. Negatively charged molecule concentrated inside the cell
- c. Threshold of excitation determines ability to respond
- d. A drop in charge inside the cell past resting potential
- e. An increase in charge inside of the cell

21. _____ All-or-none response

22. _____ Hyperpolarization

23. _____ Anion

24. _____ Polarization

25. _____ Depolarization

26 Place the steps of action potential in order from 1 to 11:

- _____ Potassium gates close
- _____ Pumps restore polarization
- _____ Cell depolarizes
- _____ Sodium rushes in
- _____ Potassium flows out

- _____ Sodium gates open
- _____ Sodium gates close
- _____ Potassium gates open
- _____ Ions return to initial concentrations
- _____ Cell repolarizes
- _____ Cell is hyperpolarized

27 What is the role of Ca^{2+} ions in nervous communication?

- a. Repolarize the cell during stimulation
- b. Exits the cell to depolarize it during impulse transmission
- c. Enters the cell to depolarize it during action potential
- d. Repolarize the cell to activate the synaptic knob
- e. Enters the cell to trigger the release of neurotransmitters

28 True or False: Neurotransmitters bind to receptors on the dendrite opening channels for positive ions to rush in and depolarize the cell.

29 Explain the difference between impulse and action potential.

Minding the Central Nervous System

Together, the brain and the spinal cord make up the central nervous system. You get the scoop on both parts in the following sections.

The spinal cord



REMEMBER The spinal cord has two primary functions: It conducts nerve

impulses to and from the brain, and it processes sensory information to produce a spinal reflex without initiating input from the higher brain centers.

The spinal cord begins at the brainstem and extends down into the lumbar region of the vertebral column. In an adult, the 18-inch spinal cord ends between the first and second lumbar vertebrae, roughly where the last ribs attach. Its tapered end is called the *conus medullaris*. The cord continues as separate strands below that point and is referred to as the *cauda equina* (horse tail). A thread of fibrous tissue called the *filum terminale* extends to the base of the *coccyx* (tailbone).

The spinal cord is an oval-shaped cylinder with two grooves running its length — the shallow *posterior median sulcus* at the back and the deeper *anterior median fissure* on the front. Also packed into the spinal cavity are the *meninges*, cerebrospinal fluid, a cushion of fat, and various blood vessels.

Three membranes called *meninges* envelop the central nervous system (brain and spinal cord), separating it from the bony cavities.

- » **The dura mater:** The outer layer, the *dura mater*, is the hardest, toughest, and most fibrous layer and is composed of white collagenous and yellow elastic fibers. It also contains numerous nerves, blood vessels, and lymphatic vessels.
- » **The arachnoid:** The middle membrane, the *arachnoid*, forms a weblike layer just inside the *dura mater* and does not contain blood vessels.
- » **The pia mater:** A thin inner membrane, the *pia mater*, lies along the surface of the central nervous system and like the *dura*, contains many blood vessels and nerves.

Between the arachnoid and the pia is the *subarachnoid space* which contains the cerebrospinal fluid (CSF) secreted by structures called *choroid plexuses*. The CSF flows through this space and circulates through the brain's ventricles, carrying nutrients pulled from the blood stream and wastes to be returned to it or into the lymphatic vessels.

Two types of solid material make up the inside of the cord, which you can see in [Figure 9-2](#): *gray matter* (which is indeed grayish in color) containing unmyelinated neurons, dendrites, cell bodies, and neuroglia; and *white matter*

(*funiculus*), so-called because of the whitish tint of its myelinated nerve fibers. At the cord's midsection is a small *central canal* filled with CSF and surrounded first by gray matter in the shape of the letter H and then by white matter, which fills in the areas around the H pattern. The top, or skinny, legs of the H are the *posterior horns* and the bottom, thicker sections are the *anterior horns*. The cross bar is the *gray commissure* which houses the central canal.

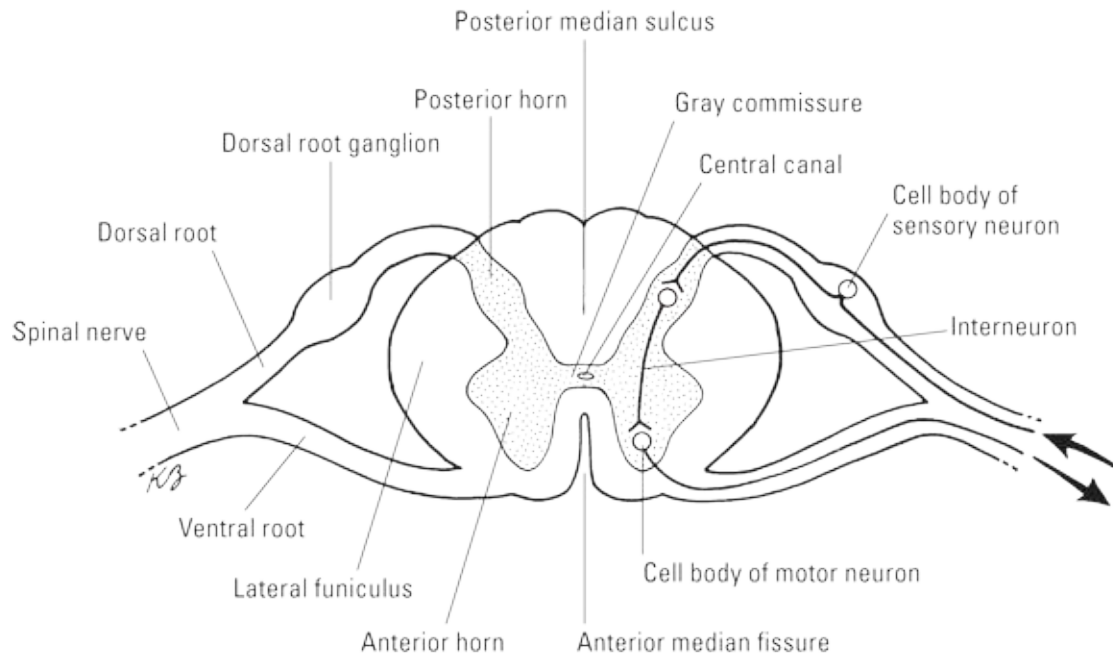


Illustration by Kathryn Born, MA

FIGURE 9-2: A cross-section of the spinal cord, showing spinal nerve connections.

The white matter consists of thousands of myelinated nerve fibers arranged in three *funiculi* (columns) on each side of the spinal cord that convey information up and down the cord's tracts. Like the gray matter (which holds the cell bodies of the axons found in the white matter), the columns are named for their location: posterior (or dorsal), anterior (or ventral), and lateral (the section on the sides of the H). The funiculi contain *nerve tracts* that are ascending (transmit sensory information to the brain), descending (transmit motor information from the brain), or bidirectional.

The brain

One of the largest organs in the adult human body, the brain tips the scales at 3 pounds and packs roughly 100 billion neurons (yes, that's billion with a "b") and 900 billion supporting neuroglia cells. In this section, we review six

major divisions of the brain from the bottom up: *medulla oblongata*, *pons*, *midbrain*, *cerebellum*, *diencephalon*, and *cerebrum*. We also discuss the ventricles.

The medulla oblongata

The spinal cord meets the brain at the *medulla oblongata*, or lower half of the *brainstem*. In fact, the medulla oblongata is continuous with the spinal cord at its base (inferiorly). It's located anterior to the cerebellum and extends upwards (superior), posterior to the pons. All the afferent and efferent tracts of the cord can be found in the brainstem as part of two bulges of white matter forming an area referred to as the *pyramids*. Many of the tracts cross from one side to the other at the pyramids, which explains why the right side of the brain controls the voluntary movement of the left side of the body and vice versa.

The medulla oblongata is responsible for maintaining our essential body functions:

- » **Cardiac center:** Controls heart rate
- » **Vasomotor center:** Controls blood pressure
- » **Respiratory center:** Adjusts depth and rate of breathing

The pons

The bulge on the anterior side of the brainstem is the *pons*. The pons is like grand central station, containing (and subsequently routing) numerous different nerve tracts and nuclei (the term for bundles of cell bodies in the brain). As a result, generating a list of functions for the pons would be quite daunting. Some of its more notable jobs, though, include facilitating communication between the cerebrum and the cerebellum, maintaining the regular breathing rate, controlling facial expressions, and causing muscle relaxation during REM sleep (so that we don't act out our dreams).

Along with the pons, the medulla oblongata also forms a network of gray and white matter called the *reticular formation*, found along the most posterior wall of the brainstem. This area is responsible for arousal — making it an essential contributor to the sleep-wake cycle as well as routing sensory information to the higher, or more advanced, parts of the brain. Without the reticular formation, we would never reach (or stay in) consciousness.

The midbrain

Between the pons and the diencephalon lies the *midbrain*, or *mesencephalon*. The midbrain contains the *cerebral aqueduct*, which connects the third and fourth ventricles. (See the section “[The ventricles](#)” later in this chapter for more.) It contains the *corpora quadrigemina*, formed by four nuclei. The superior two control visual reflexes while the inferior two control auditory ones. Also in the midbrain is the *red nucleus*, which controls reflexes to maintain posture.



REMEMBER The medulla oblongata, pons, and midbrain are collectively referred to as the brainstem.

The cerebellum

The *cerebellum* is the second-largest division of the brain. It's just above and overhangs the medulla oblongata and lies just beneath the rear portion of the cerebrum. Inside, the cerebellum resembles a tree called the *arbor vitae*, or “tree of life.” The cerebellum is divided into two hemispheres by the *falx cerebelli*, which is an invagination of the dura mater. Numerous nuclei lie within each hemisphere to coordinate the incoming sensory input with the cerebellum's motor output. It receives information from proprioceptors, the eyes, and the semi-circular canals of the ears to assess the body's position and posture. It then sends out motor signals to move the body in the desired manner. It also receives input from the motor cortex of the cerebrum to coordinate our voluntary movements.

The diencephalon

The *diencephalon*, a region between the midbrain and the cerebrum, contains separate brain structures called the *thalamus* and *hypothalamus* as well as the *optic chiasma* (where the optic nerves cross over), the *mammillary bodies* (which play a role in memory recall), and the pineal gland.

» **The thalamus** is an oval structure found just below the *corpus callosum* (the C-shaped band of white matter in the center of the brain that facilitates communication between the right and left halves of the cerebrum). The thalamus is the great integrating center of the brain with the ability to correlate the impulses from tactile, pain, and gustatory

(taste) senses with motor reactions.

- » **The epithalamus** contains the *choroid plexus*, a vascular structure that produces cerebrospinal fluid. The pineal gland and olfactory centers also lie within the epithalamus, which forms the roof of the third ventricle.
- » **The hypothalamus** is a cone-shaped structure found between the thalamus and pituitary gland. It contains the control centers for sexual reflexes; body temperature; water, carbohydrate, and fat metabolism; and emotions that affect the heartbeat and blood pressure. Its direct influence on the pituitary gland makes it the bridge between the nervous and endocrine systems.

The diencephalon and the complex set of brain structures surrounding it form the *limbic system*. This is our center of emotions, arousal, and memories. Emotions related to the limbic system include fear, anger, and various emotions related to sexual behavior. Two structures in the limbic system, the amygdala and the hippocampus, play important roles in memories. The amygdala is involved in the storage of memories while the hippocampus sends memories to the appropriate parts of the cerebral hemisphere for long-term storage and retrieves them when necessary.

The cerebrum

The *cerebrum* is mostly white matter surrounded by a thin outer layer of gray matter called the *cerebral cortex*. It features folds or convolutions called *gyri*; furrows and grooves are referred to as *sulci*, and deeper grooves are called *fissures*. A longitudinal fissure separates the cerebral hemispheres into right and left halves. Each hemisphere has a set of controls for sensory and motor activities of the body. Interestingly, it's not just right-side/left-side controls that are reversed in the cerebrum; the upper areas of the cerebral cortex control the lower body activities while the lower areas of the cortex control upper-body activities in a reversal called "little man upside down."

Different functional areas of the cerebral cortex are divided into lobes which are named for the skull bones they lie beneath. They are further divided into sensory, motor, and association areas (that provide memory and reasoning).

- » **Frontal lobe:** The most advanced part of the brain. It is responsible for our highest intellectual functions such as planning, logical reasoning, morality, and social behaviors. It contains the *primary motor cortex* which

initiates voluntary movements. The frontal lobe also contains *Broca's area* which is responsible for the generation of speech.

- » **Parietal lobe:** The primary site for integration of sensory information. It is the area responsible for our conscious perception of touch, temperature, and pain. The *somatosensory cortex* is located here, as is *Wernicke's area*, which is responsible for language comprehension. The parietal lobe is also responsible for spatial awareness and maintaining attention.
- » **Temporal lobe:** The primary auditory center. It houses situational memories such as visual scenes and music. It also plays a role in understanding and producing speech.
- » **Occipital lobe:** The primary vision center. It processes all visual information and provides meaning to it by connecting with memories.

The ventricles

The meninges of the brain provide tight control over what molecules from the bloodstream are allowed into the tissues of the brain and spinal cord. As a result, the CNS circulates CSF (cerebrospinal fluid) to provide nutrients and remove wastes. Structures within the brain, the *ventricles*, allow for this. The largest of the four, the *right and left lateral ventricles*, lie within the cerebrum and create most of the CSF. The *third ventricle* is found within the diencephalon and receives CSF from the lateral ventricles. It then sends the CSF to the *fourth ventricle* via the *cerebral aqueduct*. The fourth ventricle is in the brainstem and is continuous with the central canal of the spinal cord. It also passes CSF to the subarachnoid space.

Cranial nerves

Twelve pairs of cranial nerves connect to the central nervous system via the brain (as opposed to the 31 pairs that connect via the spinal cord). Cranial nerves are identified by Roman numerals I through XII, and memorizing them is a classic test of anatomical knowledge. Check out [Table 9-2](#) for a listing of all the nerves.

Table 9-2 Cranial Nerves

Number	Name	Type	Function
I	Olfactory	Sensory	Smell
II	Optic	Sensory	Vision

III	Oculomotor	Mixed nerve	Eyeball muscles
IV	Trochlear	Mixed nerve	Eyeball muscles
V	Trigeminal	Tri means “three,” so the three types of trigeminal nerves are 1) Ophthalmic nerve: sensory nerve; skin and mucous membranes of face and head; 2) Maxillary nerve: mixed nerve; mastication; 3) Mandibular nerve: mixed nerve; mastication	Skin; mastication (chewing)
VI	Abducens	Mixed nerve	Eye movements laterally
VII	Facial	Mixed nerve	Facial expression; salivary secretion; taste
VIII	Vestibulocochlear	Sensory	Auditory nerve for hearing and equilibrium
IX	Glossopharyngeal	Mixed nerve	Taste; swallowing muscles of pharynx; salivation; gag reflex; speech
X	Vagus	Mixed nerve	Controls most internal organs (viscera) from head and neck to transverse colon
XI	Accessory	Mixed nerve	Swallowing and phonation; rotation of head and shoulder
XII	Hypoglossal	Motor nerve	Tongue movements



TIP

The first letters of each of these nerve names, in order, are OOOTTAFFVGVAH. That’s a mouthful, but students have come up with a number of memory tools to remember them. Our favorite is Old Opera Organs Trill Terrific Arias For Various Grand Victories About History.

Put your knowledge of the central nervous system to the test:



EXAMPLE

Q. The meninges’ functions are primarily

- a. immunological.
- b. supportive.
- c. protective.
- d. a and b.
- e. b and c.

A. The correct answer is b and c (supportive and protective). Yes, meninges have two functions.

30 The cerebrum is divided into two major halves, called _____, by the _____.

- a. cerebellar hemispheres, transverse fissure
- b. cerebral spheres, central sulcus
- c. cerebellar spheres, lateral sulcus
- d. cerebral hemispheres, longitudinal fissure

31 You've been in an accident, and you have injured the left side of your cerebrum. Which areas of your body would you most likely have trouble moving?

- a. The right side
- b. The lower body
- c. The upper extremities
- d. The left side

32 Where within the cerebrum does visual activity take place?

- a. Parietal lobe
- b. Temporal lobe
- c. Occipital lobe
- d. Frontal lobe

33 What, among many things, does the cerebellum do?

- a. Controls visual activity
- b. Generates associative reasoning
- c. Interprets auditory stimulation
- d. Coordinates motor control

34-38 Match the term to its description.

- a. Has the capacity to arouse the brain to wakefulness
- b. Myelinated fibers
- c. Coordinates communication between right and left brain
- d. Bundle of neuron cell bodies
- e. Unmyelinated fibers, neuron cell bodies, and neuroglia

34. _____ White matter

35. _____ Reticular formation

36. _____ Ganglion

37. _____ Corpus callosum

38. _____ Gray matter

39-43 Match the structure to its description.

- a. Bridge connecting the medulla oblongata and cerebellum
- b. Contains the centers that control cardiac, respiratory, and vasomotor functions
- c. Controls visual and auditory reflexes
- d. Controls motor coordination and refinement of muscular movement
- e. Controls sensory and motor activity of the body

39. _____ Pons

40. _____ Cerebellum

41. _____ Medulla oblongata

42. _____ Cerebrum

43. _____ Midbrain

44 What does the choroid plexus do?

- a. Filters cerebrospinal fluid
- b. Absorbs cerebrospinal fluid
- c. Produces cerebrospinal fluid
- d. Eliminates exhausted cerebrospinal fluid

45-57 Use the terms that follow to identify the parts of the brain shown in [Figure 9-3](#).

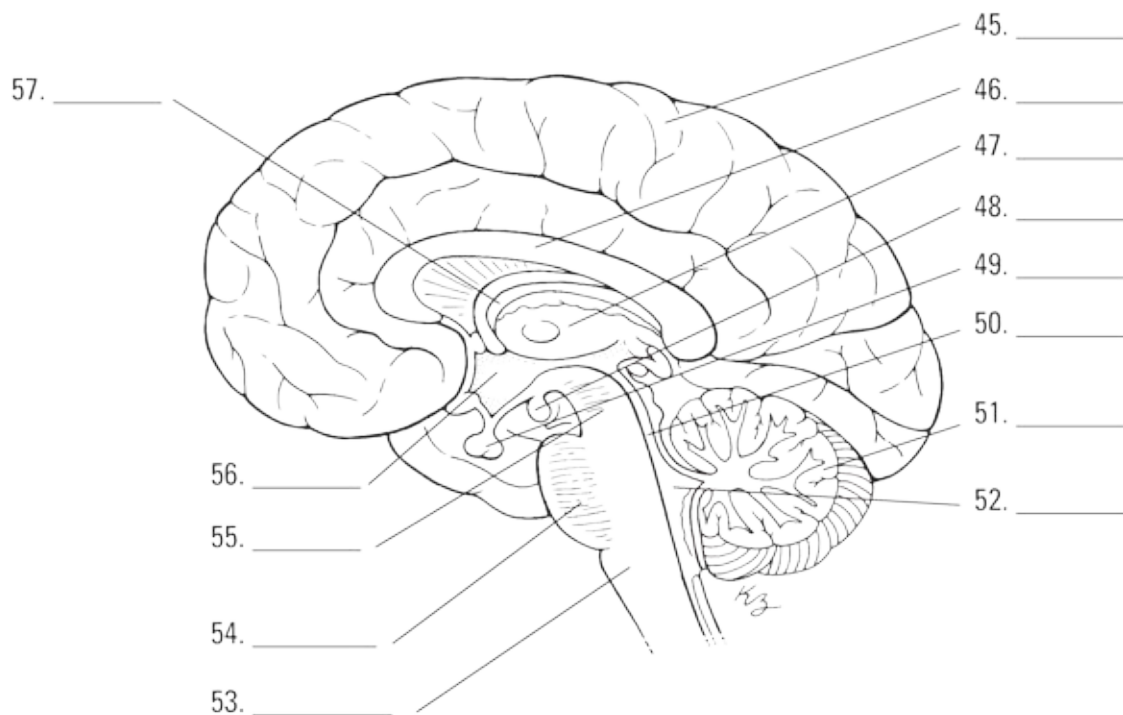


Illustration by Kathryn Born, MA

FIGURE 9-3: Sagittal view of the brain.

- a. Pons
- b. Thalamus
- c. Cerebellum
- d. Corpus callosum
- e. Third ventricle

- f. Hypothalamus
- g. Cerebrum
- h. Cerebral aqueduct
- i. Midbrain
- j. Pituitary gland
- k. Medulla oblongata
- l. Fourth ventricle
- m. Mammillary body

Taking Side Streets: The Peripheral Nervous System

The *peripheral nervous system* (PNS) is the network that carries information to and from the spinal cord. Among its key structures are 31 pairs of spinal nerves (see [Figure 9-4](#)). Also included in the PNS are all the individual neuron receptors that relay information as well as some glial cells. Eight of the spinal nerve pairs are *cervical* (having to do with the neck), 12 are *thoracic* (relating to the chest, or thorax), five are *lumbar* (between the lowest ribs and the pelvis), five are *sacral* (the posterior section of the pelvis), and one is *coccygeal* (relating to the tailbone). Spinal nerves connect with the spinal cord by two bundles of nerve fibers, or *roots*. The *dorsal root* contains afferent fibers that carry sensory information to the central nervous system. Cell bodies of sensory neurons lie outside the spinal cord in a bulging area called the *dorsal root ganglion* (refer to the cross-section of the spinal cord in [Figure 9-2](#)). This allows the spinal cord to organize incoming information routing it to the appropriate ascending nerve tract. A second bundle, the *ventral root*, contains efferent motor fibers; these cell bodies lie inside the spinal cord. In each spinal nerve, the two roots join outside the spinal cord to form what's called a *mixed spinal nerve*.

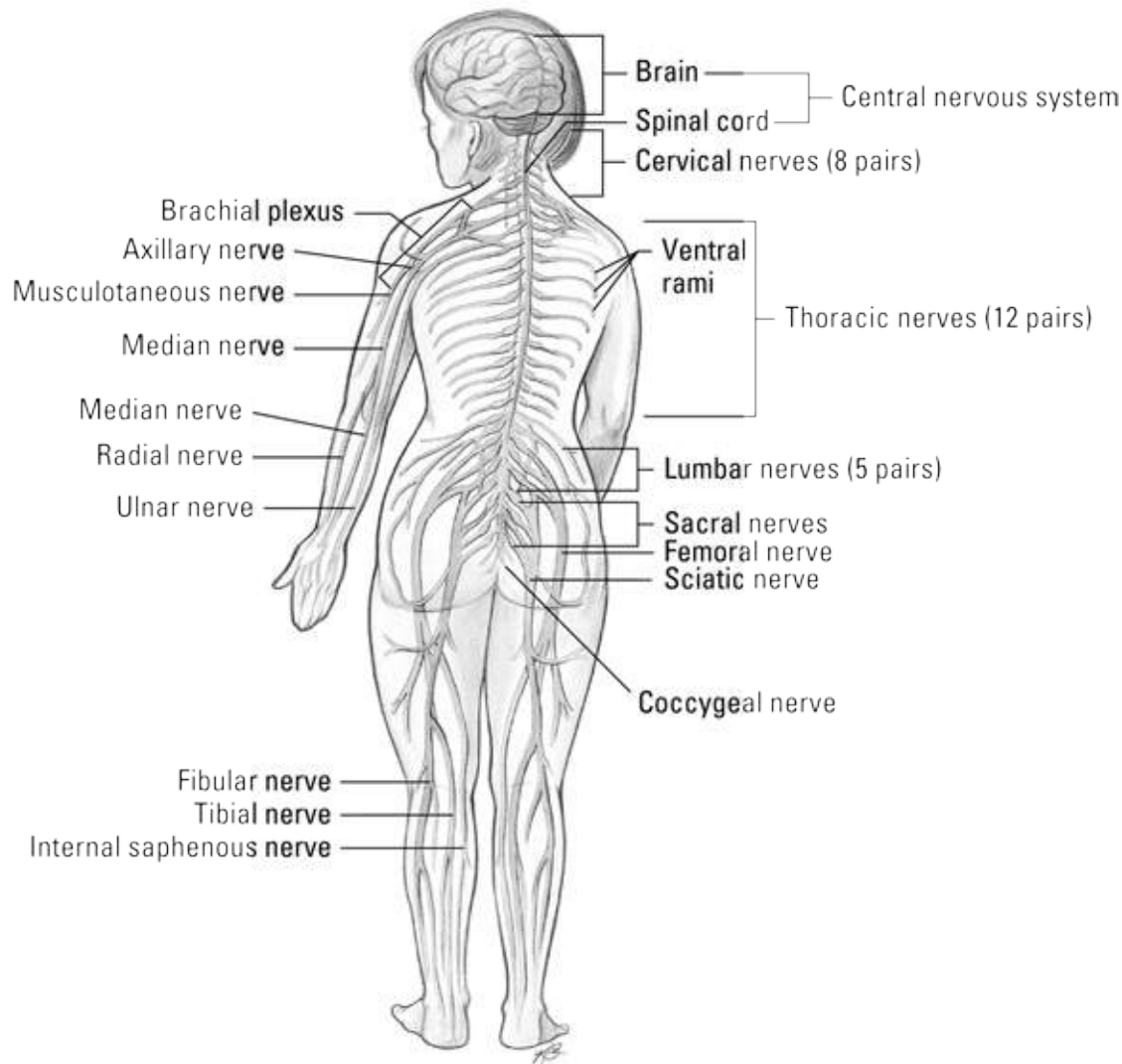


Illustration by Kathryn Born, MA

FIGURE 9-4: The spinal nerves plus branching plexus nerves.

Spinal reflexes, or *reflex arcs*, occur when a sensory neuron transmits a “danger” signal — like a sensation of burning heat — through the dorsal root ganglion. An interneuron (or association neuron) in the spinal cord receives the stimulus and passes along the signal directly to a motor neuron (or efferent fiber) that stimulates a muscle, which immediately pulls the burning body part away from heat (see [Figure 9-5](#)). Other reflexes, like the knee-jerk, or *patellar reflex*, forego the interneuron and the sensory one directly stimulates the motor one.

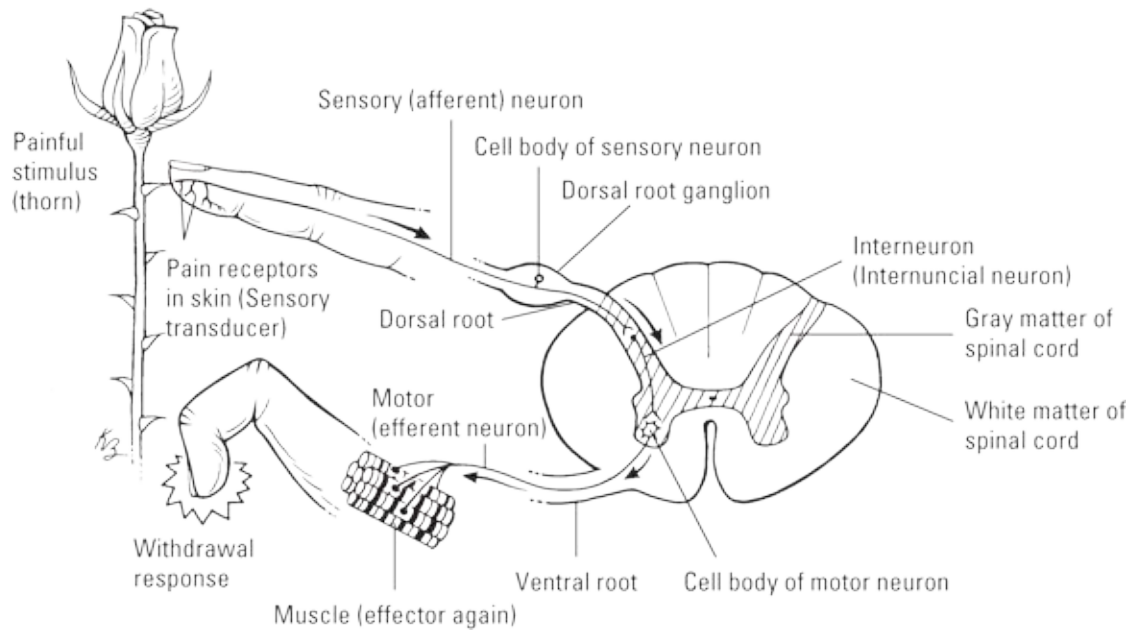


Illustration by Kathryn Born, MA

FIGURE 9-5: A reflex arc — responding to pain.

After a spinal nerve leaves the spinal column, it divides into two small branches. The posterior, or *dorsal ramus*, goes along the back of the body to supply a specific segment of the skin, bones, joints, and longitudinal muscles of the back. The ventral, or *anterior ramus*, is larger than the dorsal ramus and supplies the anterior and lateral regions of the trunk and limbs.

Groups of spinal nerves interconnect to form an extensive network called a *plexus* (Latin for “braid”), each of which connects through the anterior ramus, including the *cervical plexus* of the neck, *brachial plexus* of the shoulder and axilla, and *lumbosacral plexus* of the lower back (including the body’s largest nerve, the *sciatic nerve*). However, there’s no plexus in the thoracic area. Instead, the anterior ramus directly supplies the intercostal muscles (literally “between the ribs”) and the skin of the region.

58 What enters the nervous system through a dorsal root ganglion?

- A dorsal ramus
- An anterior ramus
- An efferent nerve
- An afferent nerve

- e. The lumbosacral plexus

59 What is a mixed spinal nerve?

- a. The point outside the spinal cord where the dorsal root and the ventral root join
- b. The bulging area inside the spinal cord where afferent fibers deliver sensory information
- c. A neuromere that interacts with a plexus
- d. The point along the spinal cord where the dorsal ramus and the anterior ramus join
- e. A nerve that runs in and then directly out of the spinal cord creating a reflex arc

60 Networks of nerves often form a plexus, from the Latin word for “braid.” Which spinal nerve area does not?

- a. Cervical
- b. Lumbar
- c. Sacral
- d. Thoracic

Keep Breathing: The Autonomic Nervous System

Just as the name implies, the autonomic nervous system functions automatically. Divided into the *sympathetic* and *parasympathetic* systems, it activates the involuntary smooth and cardiac muscles and glands to carry out all the body functions that happen automatically. (Autonomic, automatic — see the connection?) Autonomic functions are under the control of the hypothalamus, cerebral cortex, brainstem.

The sympathetic system, which is responsible for the body’s involuntary fight-or-flight response to stress, is defined by the autonomic fibers that exit

the thoracic and lumbar segments of the spinal cord. The parasympathetic system (affectionately referred to as “rest and digest”) is defined by the autonomic fibers that either exit the brainstem via the cranial nerves or exit the sacral segments of the spinal cord. (You can see both systems in [Figure 9-6.](#))

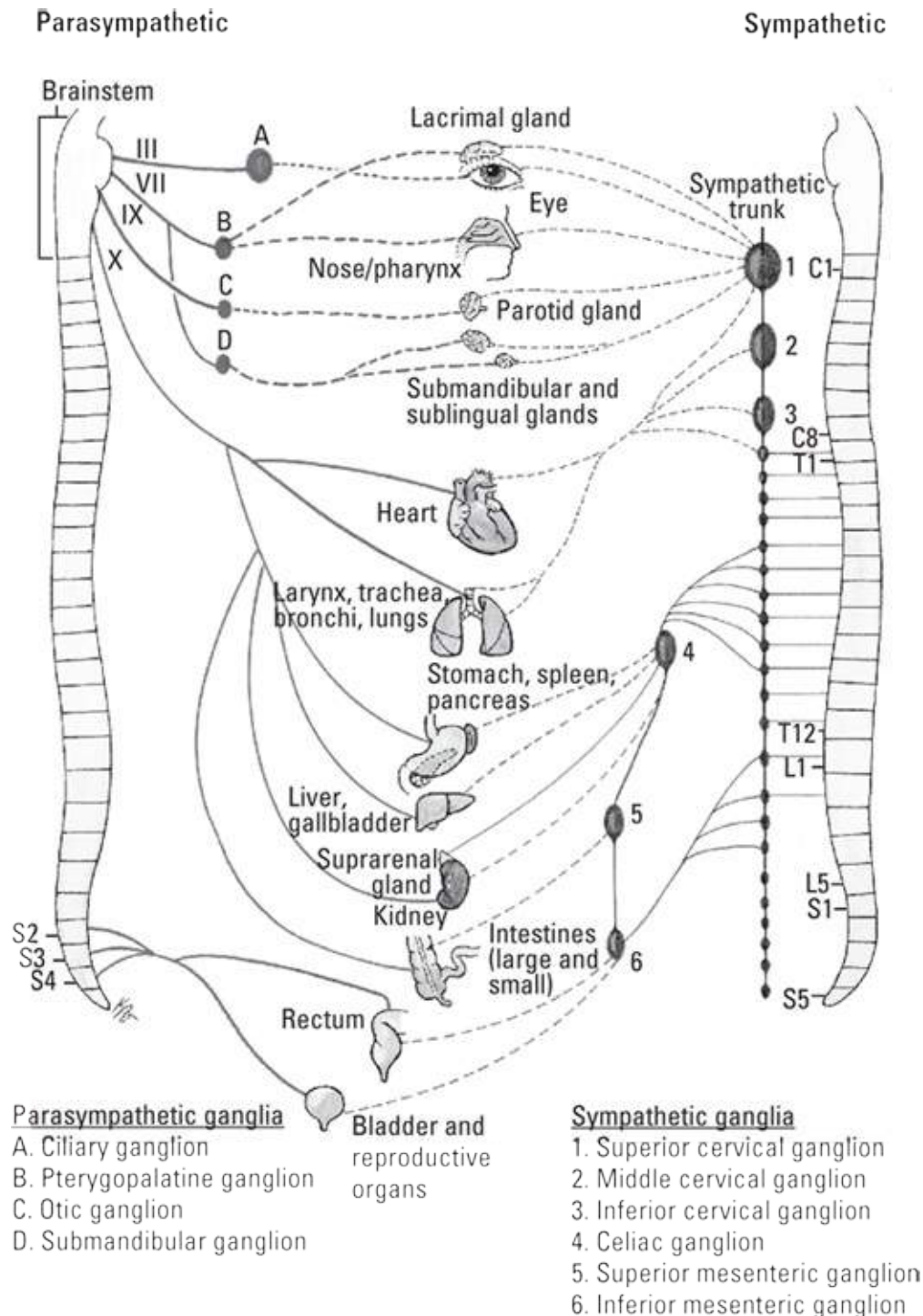


Illustration by Kathryn Born, MA

FIGURE 9-6: The sympathetic and parasympathetic nervous systems.



REMEMBER The sympathetic and parasympathetic systems oppose each other in function, helping to maintain *homeostasis*, or balanced activity in the body systems. Yet, often the sympathetic and parasympathetic systems work in concert. The sympathetic system dilates the eye's pupil, but the parasympathetic system contracts it again. The sympathetic system quickens and strengthens the heart while the parasympathetic slows the heart's action. The sympathetic system contracts blood vessels in the skin so more blood goes to muscles for a fight-or-flight reaction to stress, and the parasympathetic system dilates the blood vessels when the stress concludes.



REMEMBER The motor division of the nervous system (all outgoing information) is split into the autonomic and *somatic nervous system*. This carries out the voluntary orders for skeletal muscles to contract.

See whether any of the following practice questions touch a nerve:

- 61** Which three parts of the brain control autonomic functions?
- a. Medulla oblongata, parietal lobe, and cerebellum
 - b. Occipital lobe, cerebral cortex, and temporal lobe
 - c. Hypothalamus, cerebral cortex, and brainstem
 - d. Brainstem, Broca's area, and frontal lobe
 - e. Medulla oblongata, pons, and hypothalamus
- 62** Which of the following statements is true about the autonomic nervous system?
- a. It has two parts: the parasympathetic that carries out the stress response, and the sympathetic that controls all normal functions.

- b. It's the nervous system that controls all reflexes.
- c. It doesn't function when the body's under stress.
- d. It has two divisions that are antagonistic to each other, meaning that one counteracts the effects of the other one.
- e. It controls the voluntary override for the body's autonomic actions.

63 What is the primary function of the autonomic nervous system?

- a. It activates in response to emotional stress.
- b. It controls involuntary body functions.
- c. It's the primary source of sensory input to the body.
- d. It controls the contraction of skeletal muscles.
- e. It works to block signals that may overload the overall nervous system.

Coming to Your Senses

The nervous system must have some way to perceive its environment in order to generate appropriate responses. That's where the senses come in. *Sense receptors* are the specialized nervous cells that respond to stimuli — like increased temperature, bitter tastes, and sharp points — by generating a nerve impulse. Although there are millions of *general sense receptors* found throughout the body that can convey touch, pain, and physical contact, there are far fewer of the *special sense receptors* — those located in the head — that really bring meaning to your world. Here we discuss vision, hearing, and equilibrium. The special senses of olfaction (smell) and gustation (taste) are discussed in [Chapters 13](#) and [14](#), respectively.



REMEMBER Sense receptors are classified both by the location of the stimulus as well as their mechanism of action. With respect to location, these cells can be *exteroceptors*, which respond to stimulus on the body surface; *interoceptors*, which monitor internal stimuli; and *telereceptors*, which respond to stimuli without physical contact (sight, hearing, smell). [Table 9-3](#) shows the major categories of receptors based on the type of

stimulus they respond to.

Table 9-3 Sensory Receptors

Receptor Type	Stimulus	Example
Chemoreceptor	Change in chemical concentration	Olfactory, pH
Nociceptor	Tissue damage	pain
Thermoreceptor	Temperature change	Hot and cold
Mechanoreceptor	Pressure/mechanical forces	Touch, hearing
Photoreceptor	Light	Rods and cones
Proprioceptor	Stretch	Muscle spindles

On watch: The eyes

Although there are many romantic notions about eyes, the truth is that an eyeball is simply a hollow sphere bounded by a trilayer wall and filled with a gelatinous fluid called, oddly enough, *vitreous humor*. The outer fibrous coat is made up of the *sclera* (the white of the eye) and the *cornea* in front. The sclera provides mechanical support, protection, and a base for attachment of eye muscles, which assist in the focusing process. The cornea covers the anterior with a clear window.

An intermediate, or vascular, layer called the *uvea* provides blood and lymphatic fluids to the eye; regulates the amount of light entering the eye; and secretes and reabsorbs *aqueous humor*, a thin, watery liquid that fills the anterior chamber of the eyeball in front of the iris and the posterior chamber between the iris and the lens. The uvea has three components:

- » **The iris:** Contains blood vessels and smooth muscle fibers to control the pupil's diameter. The pigments that provide our eye color prevent light from entering anywhere other than the pupil.
- » **The choroid coat:** A thin, dark brown, vascular layer lining most of the sclera on the back and sides of the eye. The choroid contains arteries, veins, and capillaries that supply the *retina* and sclera with nutrients, and it also contains pigment cells to absorb light and prevent reflection and blurring.
- » **The ciliary body:** An extension of the choroid coat that leads to the iris. It also attaches to the lens via *suspensory ligaments* and controls the lens'

shape to aid in focusing.

The crystalline lens consists of concentric layers of protein. It's *biconvex* in shape, bulging outward and with help from the cornea, bends light projecting it into the *inner tunic*. When the muscles of the ciliary body contract, the shape of the lens changes, altering the visual focus. This process of *accommodation* allows the eye to see objects both at a distance and close-up.

The inner tunic is the internal lining of the eye. Here, the *retina* lies atop the choroid coat containing the photoreceptors that will communicate with the occipital lobe of the brain. The space created, called the *posterior cavity*, is filled with *vitreous humor*, a jelly-like fluid that maintains the shape of the eye. The photoreceptors of the retina come in two varieties: *rods* and *cones*, whose axons are bundled into the optic nerve that exits out the back. The rods are dim light receptors that provides the general shape of the image. The cones detect bright light and provide sharpness and color to the image. The retina has an *optic disc*, which is essentially a blind spot incapable of producing an image. The eye's blind spot is a result of the absence of photoreceptors in the area of the retina where the optic nerve leaves the eye.

The macula or *macula lutea* (Latin for "yellow spot") is an oval-shaped, highly pigmented yellow area near the center of the retina. There is a small dimple near the center of the macula called the *fovea centralis* (*fovea* is Latin for "pit"). This is where the eye's vision is sharpest and it is where most of the eye's color perception occurs. Only cones can be found in this area, smaller and more closely packed than anywhere else in the retina. The fovea sees only the central 2 degrees of the visual field, which encompasses roughly twice the width of your thumbnail at arm's length. If an observed area or object is larger than this angle of vision, such as when reading, the eyes must constantly shift their gaze, a process known as *saccadic movement*, to bring different portions of the image into the fovea. For this reason, the cognitive brain is responsible for oculomotor commands and piecing together a comprehensive visual image.

The *palpebrae* (eyelids) extend from the edges of the eye orbit, into which roughly five-sixths of the eyeball is recessed. A mucous membrane called the *conjunctiva* covers the inner surface of each eyelid and the anterior surface of the eye, but not the cornea. (The infamous "pink eye" is the inflammation of this membrane.) Up top and to the side of the orbital cavities are lacrimal

glands that secrete tears that are carried through a series of lacrimal ducts to the conjunctiva of the upper eyelid. Ultimately, secretions drain from the eyes through the nasolacrimal ducts.

Listen up: The ears

Human ears — otherwise called *vestibulocochlear* organs — are more than just organs of hearing. They also serve as organs of equilibrium, or balance. Here are the three divisions of the ear:

- » **The external ear** includes the *auricle*, or *pinna*, which is the recognizable folded appendage made of elastic cartilage and skin. Extending into the skull is the *ear canal*, or *external auditory meatus*, a short passage through the temporal bone ending at the *tympanic membrane*, or *eardrum*. Sebaceous glands near the external opening and *ceruminous* glands in the upper wall produce the brownish substance known as earwax, or *cerumen*.
- » **The middle ear** is a small, usually air-filled cavity in the skull that's lined with mucous membrane. It contains the *Eustachian tube* that connects with the pharynx to keep air pressure equal on both sides of the tympanic membrane. Three small bones called *auditory ossicles* occupy the middle ear, deriving their names from their shapes: the *malleus* (hammer), the *incus* (anvil), and the *stapes* (stirrup).
- » **The internal ear** has three main components: the *semi-circular canals*, the *cochlea*, and the *vestibule*, which is the space between them. The cochlea is the snail-shaped structure that contains the *hair cells* — the mechanoreceptors that transmit auditory input to the temporal lobe. The semi-circular canals are three fluid-filled tubes that communicate information about the body's position to the cerebellum via the vestibulocochlear nerve. When we move, tilting to one side for example, the fluid inside does not. This stimulates the proprioceptors lining the canals.



REMEMBER The process of hearing a sound follows these basic steps:

1. Sound waves travel through the auditory canal, striking the eardrum,

making it vibrate.

2. This vibration is passed through the malleus, the incus, and finally the stapes.
3. The stapes then vibrates the oval window of the vestibular canal, translating the motion into the *perilymph fluid* of the cochlea.
4. The vibrating fluid begins stimulating the *endolymph fluid* in the membranous area of the cochlea.
5. The endolymph fluid in turn stimulates the hair cells, which transmit the impulses to the brain over the auditory (cochlear) nerve.



EXAMPLE Q. The most sensitive region of the retina producing the greatest visual acuity is the

- a. optic disc.
- b. cornea.
- c. fovea centralis.
- d. choroid coat
- e. macula lutea.

A. The correct answer is fovea centralis. It's loaded with light-sensitive cones.

64-77 Use the terms that follow to identify the internal structures of the eye shown in [Figure 9-7](#).

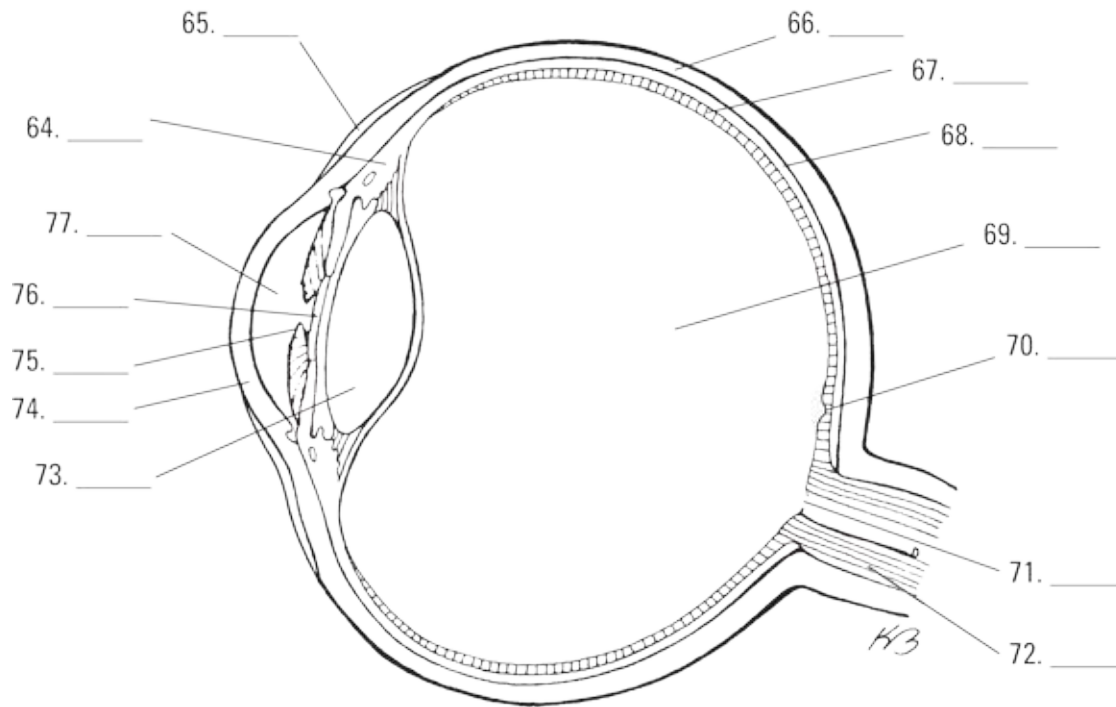


Illustration by Kathryn Born, MA

FIGURE 9-7: The internal structures of the eye.

- a. Optic disc
- b. Pupil
- c. Optic nerve
- d. Retina
- e. Cornea
- f. Sclera
- g. Ciliary body
- h. Fovea centralis
- i. Lens
- j. Anterior chamber (contains aqueous humor)
- k. Choroid coat
- l. Conjunctiva
- m. Iris
- n. Posterior cavity (contains vitreous humor)

78 Outline the path of light through the eye.

79 The accommodation, or focusing, of the eye involves which of these?

- a. Sphincter of the pupil
- b. Contraction of the iris
- c. Action of the ciliary muscles
- d. Adjustment of the cornea
- e. Contraction of the pupil

80 All the ganglion cells of the retina merge and exit the eye forming the optic nerve. What is the area where they exit called?

- a. Macula lutea
- b. Optic disc
- c. Choroid coat
- d. Vitreous humor
- e. Fovea centralis

81-88 Sound waves travel through the ear canal and cause the **81.**____, or eardrum, to vibrate. Those vibrations then are picked up by three small bones, called **82.**____. The **83.**____ is the first to receive the vibrations. It then passes them to the **84.**____ and then the **85.**____. These vibrations are then transferred to the **86.**____, where fluid is jostled stimulating the **87.**____. These communicate the sounds to the auditory cortex in the **88.**____ of the brain.

- a. Acoustic meatus

- b. Cochlea
- c. Hair cells
- d. Incus
- e. Malleus
- f. Occipital lobe
- g. Ossicles
- h. Semicircular canals
- i. Stapes
- j. Temporal lobe
- k. Tympanic membrane
- l. Vestibule

89 What is the difference between aqueous humor and vitreous humor?

- a. One lines the sclera, and the other bathes the iris.
- b. One is thin and watery, and the other is thick and gelatinous.
- c. One acts as a photoreceptor, and the other acts as a focusing mechanism.
- d. One is subtly funny, and the other is brashly obvious.

90 What is the role of the semicircular canals?

- a. provide structure and support to the orbit
- b. contain rods, cones, and hair cells to receive stimulus
- c. translate vibration of the ossicles to sound
- d. equalize pressure in the ear's chambers
- e. transmit information on the body's position

Answers to Questions on the Nervous System

The following are answers to the practice questions presented in this chapter.

- 1-5 Fill in the blanks. The communicating cells of the nervous system, or **1. neurons** have two essential properties. First is **2. irritability**, the tendency to respond to a stimulus. Second is the ability to generate electricity, called **3. conductivity**. This flow of electricity is known as **4. impulse** but cannot be used to communicate with other neurons. For this, **5. neurotransmitters** are used.
- 6 What are the three primary functions of the nervous system? **c. Perception, integration, action**. Each of the other answer options includes at least one function that clearly belongs to another of the body's systems.
- 7 Which structure is found at the end of an axon? **e. synaptic knob**. Though you could argue that soma is true as well, but that's not really the end of the axon, that's the beginning.
- 8 Which part of a neuron is responsible for conducting electrical signals toward the cell body? **d. dendrite**
- 9 Which membrane surrounds an axon (nerve fiber)? **b. Neurilemma**. It actually wraps around the myelin sheath, so it's always on the outside.
- 10 Astrocytes: **f. Cells that contribute to the repair process of the central nervous system**.
- 11 Microglia: **d. Cells that are phagocytic**.
- 12 Oligodendrocytes: **b. Cells that form and preserve myelin sheaths in CNS**.
- 13 Schwann cell: **a. Cells that form and preserve myelin sheaths in PNS**.
- 14 Satellite cell: **c. Cells that provide nutrients to bundled cell bodies**.
- 15 Ependyma: **e. Cells that form linings in the CNS**.
- 16 How many dendrites and axons are there on a multipolar neuron? **a. Many dendrites and one axon**. A neuron may have one, many, or no dendrites, but it always has a single axon.

- 17) A synapse between neurons is best described as the transmission of **d. an impulse through a chemical change**. With all that acetylcholine and cholinesterase floating around, it must be a chemical transmission.
- 18) The connective tissue that surrounds a bundle of fibers in a nerve is known as the **b. perineurium**.
- 19) Nervous tissue is essentially neurons connected end-to-end with very little matrix. **False**. Abundant space is visible between neurons, dotted with glial cells and numerous “fibers” which are actually axons of other neurons.
- 20) Which statement best describes how neurons communicate? **d. In response to stimulus, the dendrites generate an electrical signal which is passed to the cell body. The axon will then spread the impulse down its length releasing a chemical at the end**. Note that the electrical signal in a neuron is not termed impulse until it is in the axon.
- 21) All-or-none response: **c. Threshold of excitation determines ability to respond**
- 22) Hyperpolarization: **d. A drop in charge inside the cell past resting potential**
- 23) Anion: **b. Negatively charged molecule concentrated inside the cell**
- 24) Polarization: **a. A differential in charge from outside of the cell to the inside**
- 25) Depolarization: **e. An increase in charge inside of the cell**
- 26) Place the steps of action potential in order: **8. Potassium gates close, 10. Pumps restore polarization, 3. Cell depolarizes, 2. Sodium rushes in, 6. Potassium flows out, 1. Sodium gates open, 4. Sodium gates close, 5. Potassium gates open, 11. Ions return to initial concentrations, 7. Cell repolarizes, 9. Cell is hyperpolarized**. Though steps 4 and 5 happen simultaneously (Na^+ gates open, K^+ gates close), it makes more sense to address the open sodium gates first.
- 27) What is the role of Ca^{2+} ions in nervous communication? **e. Enters the**

cell to trigger the release of neurotransmitters

- 28) Neurotransmitters bind to receptors on the dendrite opening channels for positive ions to rush in and depolarize the cell: **False**. This is a bit of a trick question because that's exactly what excitatory neurotransmitters do. But there are also inhibitory neurotransmitters which can lead to hyperpolarization instead.
- 29) Explain the difference between impulse and action potential . **Impulse is the step-wise process that occurs in a single segment of axon. It begins with voltage-gated sodium channels opening, leading to local depolarization, and ends with the sodium/potassium pumps returning the ions to their original positions. However, as the sodium ions rush in and depolarize that segment of axon, the threshold is reached in the next segment, opening the next set of sodium channels. So the steps early in each action potential triggers the start of the next one. This progression of action potentials constitutes the impulse.**
- 30) The cerebrum is divided into two major halves, called **d. cerebral hemispheres**, by the **d. longitudinal fissure**. Cerebrum = cerebral, and two halves = hemispheres. You can remember the fissure's name by equating it to Earth's prime meridian, which separates the Eastern and Western Hemispheres, and longitudinal is the most likely position for an equal division.
- 31) You've been in an accident, and you have injured the left side of your cerebrum. Which areas of your body would you most likely have trouble moving? **a. The right side**. In the cerebrum, right = left and up = down. Clear as mud?
- 32) Where within the cerebrum does visual activity take place? **c. Occipital lobe**. To remember, use the word "occipital" to bring to mind the word "optic," which of course is related to visual activity.
- 33) What, among many things, does the cerebellum do? **d. Coordinates motor control**. As the second-largest division of the brain, the cerebellum also is known as the "little brain."
- 34) White matter: **b. Myelinated fibers**.

- 35) Reticular formation: **a. Has the capacity to arouse the brain to wakefulness.**
- 36) Ganglion: **d. Bundle of neuron cell bodies.**
- 37) Corpus Callosum: **c. Coordinates communication between right and left brain.**
- 38) Gray matter: **e. Unmyelinated fibers, neuron cell bodies, and neuroglia.**
- 39) Pons: **a. Bridge connecting the medulla oblongata and cerebellum.**
- 40) Cerebellum: **d. Controls motor coordination and refinement of muscular movement.**
- 41) Medulla oblongata: **b. Contains the centers that control cardiac, respiratory, and vasomotor functions.**
- 42) Cerebrum: **e. Controls sensory and motor activity of the body.**
- 43) Midbrain: **c. Controls visual and auditory reflexes.**
- 44) What does the choroid plexus do? **c. Produces cerebrospinal fluid.** This one requires rote memorization — sorry!
- 45-57) Following is how [Figure 9-3](#), the brain, should be labeled:
45. **g. Cerebrum**; 46. **d. Corpus callosum**; 47. **b. Thalamus**; 48. **m. Mammillary body**; 49. **j. Pituitary gland**; 50. **h. Cerebral aqueduct**; 51. **c. Cerebellum**; 52. **l. Fourth ventricle**; 53. **k. Medulla oblongata**; 54. **a. Pons**; 55. **i. Midbrain**; 56. **f. Hypothalamus**; 57. **e. Third ventricle.**
- 58) What enters the nervous system through a dorsal root ganglion? **d. an afferent nerve.** The dorsal root brings in sensory information. Besides, all the other terms are efferent structures.
- 59) What is a mixed spinal nerve? **a. The point outside the spinal cord where the dorsal root and the ventral root join.** That juncture occurs in each of the 31 pairs of spinal nerves.

- 60) Networks of nerves often form a plexus, from the Latin word for “braid.” Which spinal nerve area does not? **d. Thoracic.** There is far less nerve networking to be “braided” there.
- 61) Which three parts of the brain control autonomic functions? **c. Hypothalamus, cerebral cortex, and brainstem.** We may have thrown you for a loop by calling the medulla oblongata and the pons by their more common name: brainstem. Beware of Choice e: While associated with advanced functions, the cerebral cortex plays a role here too.
- 62) Which of the following statements is true about the autonomic nervous system? **d. It has two divisions that are antagonistic to each other, meaning that one counteracts the effects of the other one.** As a result, the body achieves homeostasis.
- 63) What is the primary function of the autonomic nervous system? **b. It controls involuntary body functions.** That’s the only answer option with a sense of automation.
- 64-77) Following is how [Figure 9-7](#), the internal structures of the eye, should be labeled.

64. **g. Ciliary body**; 65. **l. conjunctiva**; 66. **f. Sclera**; 67. **d. Retina**; 68. **k. Choroid coat**; 69. **n. Posterior cavity (filled with vitreous humor)**; 70. **h. Fovea centralis**; 71. **a. Optic disc**; 72. **c. Optic nerve**; 73. **i. Lens**; 74. **e. Cornea**; 75. **m. Iris**; 76. **b. Pupil**; 77. **j. Anterior chamber (contains aqueous humor).**

- 78) Outline the path of light through the eye. **Cornea, anterior chamber, pupil, posterior chamber, lens, posterior cavity, retina.**



TIP You may want to develop your own mnemonic for this path. Here’s an unwieldy example: Continuous Air Conditioning Puts Possible Charges Lower on the Politically Correct Radar.

- 79) The accommodation, or focusing, of the eye involves which of these? **c. Action of the ciliary muscles.** They reshape the lens by contracting and

relaxing as needed to bring things into focus.

80) All the ganglion cells of the retina merge and exit the eye forming the optic nerve. What is the area where they exit called? **b. Optic disc.** Sometimes referred to as the *optic nerve head*, the optic disc has no photoreceptors of its own, causing a “blind spot” in each of your eyes.

81-88) Sound waves travel through the ear canal and cause the **k. tympanic membrane**, or eardrum, to vibrate. Those vibrations then are picked up by three small bones, called **g. ossicles**. The **e. malleus** is the first to receive the vibrations. It then passes them to the **d. incus** and then the **i. stapes**. These vibrations are then transferred to the **b. cochlea**, where fluid is jostled stimulating the **c. hair cells**. These communicate the sounds to the auditory cortex in the **j. temporal lobe** of the brain.



TIP

A mnemonic for the ossicles can be helpful here: Mailing Includes Stamps.

89) What is the difference between aqueous humor and vitreous humor? **b. One is thin and watery, and the other is thick and gelatinous.** Vitreous humor (which is thick and gelatinous) fills the hollow sphere of the eyeball, but aqueous humor (which is thin and watery) fills two chambers toward the front of the eyeball.

90) What is the role of the semicircular canals? **e. transmit information on the body's position**

Chapter 10

Raging Hormones: The Endocrine System

IN THIS CHAPTER

- » Absorbing what endocrine glands do
 - » Checking out the ringmasters: The pituitary and hypothalamus glands
 - » Surveying the supporting glands
 - » Understanding how the body balances under stress
-

The human body has two separate command and control systems that work in harmony most of the time but also work in very different ways. Designed for instant response, the nervous system cracks its cellular whip using electrical signals that make entire systems hop to their tasks with no delay (you can get a feel for the nervous system in [Chapter 9](#)). By contrast, the endocrine system's glands use chemical signals called *hormones* that behave like the steering mechanism on a large, fully loaded ocean tanker; small changes can have big impacts, but it takes quite a bit of time for any evidence of the change to make itself known. At times, parts of the nervous system stimulate or inhibit the secretion of hormones, and some hormones are capable of stimulating or inhibiting the flow of nerve impulses.

The word “hormone” originates from the Greek word *hormao*, which literally translates as “I excite.” And that’s exactly what hormones do. Each chemical signal stimulates specific parts of the body, known as *target tissues* or *target cells*. The body needs a constant supply of hormonal signals to grow, maintain homeostasis, reproduce, and conduct myriad processes.

In this chapter, we go over which glands do what and where, as well as review the types of chemical signals that play various roles in the body. You also get to practice discerning what the endocrine system does, how it does it, and why the body responds like it does.

No Bland Glands



REMEMBER Technically, there are ten or so primary endocrine glands, made up of epithelial cells embedded within connective tissue. There are also various other hormone-secreting tissues scattered throughout the body. *Exocrine glands*, like the sweat and mammary glands, release their secretions through ducts destined for a body surface. (Keep in mind your body has both external and internal surfaces.) Like endocrine glands, the *paracrine glands* secrete their product into the interstitial fluid. The *paracrine factors* then bind to receptors on nearby cells inducing their response. Secretions from the endocrine glands, though, are pulled into capillaries for destinations elsewhere in the body. So our hormones make the journey to their target cells via the bloodstream without errantly triggering an unrelated event along the way. This is due to their structural specificity; that is, each different hormone has a unique shape that seems tailor-made for the receptors of its target cells.

Hormones can be classified either as *steroid* (derived from cholesterol) or *nonsteroid* (derived from amino acids and other proteins). The steroid hormones — which include testosterone, estrogen, and cortisol — are the ones most closely associated with emotional outbursts and mood swings. Because they are lipid-soluble, steroid hormones can enter just about any cell in the body. However, to trigger the desired response, a receptor molecule inside of the cell must bind the hormone and carry it to the nucleus.

Nonsteroid hormones are divided among four classifications:

- » Some are *modified amino acids*, including such things as epinephrine and norepinephrine, as well as melatonin.
- » Others are *peptide*-based (at least three amino acids), including antidiuretic hormone (ADH) and oxytocin (OT).
- » *Glycoprotein*-based hormones include follicle-stimulating hormone (FSH), and luteinizing hormone (LH) — both closely associated with the reproductive systems (which we cover in [Chapters 16](#) and [17](#)).

» *Protein*-based nonsteroid hormones include such crucial substances as insulin and growth hormone (GH) as well as prolactin and parathyroid hormone.

Since these hormones are water-soluble, they are unable to enter the target cell. Instead, they bind to receptors on the cell's surface triggering a cascade of chemical reactions using a *second messenger* inside of the cell, ultimately leading to the desired effect.



REMEMBER Hormone functions include controlling the body's internal environment by regulating its chemical composition and volume, activating responses to changes in environmental conditions to help the body cope, influencing growth and development, enabling several key steps in reproduction, regulating components of the immune system, and regulating organic metabolism.

See if all this hormone-speak is sinking in:

1-5 Mark the statement with a T if it's true or an F if it's false:

1. _____ The endocrine system brings about changes in the metabolic activities of the body tissue.
2. _____ The amount of hormone released is determined by the body's need for that hormone at the time.
3. _____ The glands of the endocrine system are composed of cartilage cells.
4. _____ Endocrine glands aren't functional in reproductive processes.
5. _____ Some hormones can be derivatives of amino acids, whereas others are synthesized from cholesterol.

6 _____ glands secrete their product through ducts while _____ glands secrete their product into the interstitial fluid, which flows into the blood.

- a. Exocrine; endocrine

- b. Endocrine; exocrine
- c. Endocrine; paracrine
- d. Paracrine, heterocrine
- e. Exocrine; paracrine

7 Explain how, due to their structural differences, steroid and nonsteroid hormones elicit responses from their target cells in different ways.

Enter the Ringmasters

The key glands of the endocrine system include the *pituitary* (also called the *hypophysis*), *adrenal* (also referred to as *suprarenal*), *thyroid*, *parathyroid*, *thymus*, *pineal*, *islets of Langerhans* (within the *pancreas*), and *gonads* (testes in the male and ovaries in the female). But of all these, it's the pituitary working in concert with the hypothalamus in the brain that really keeps things rolling (see [Figure 10-1](#)).

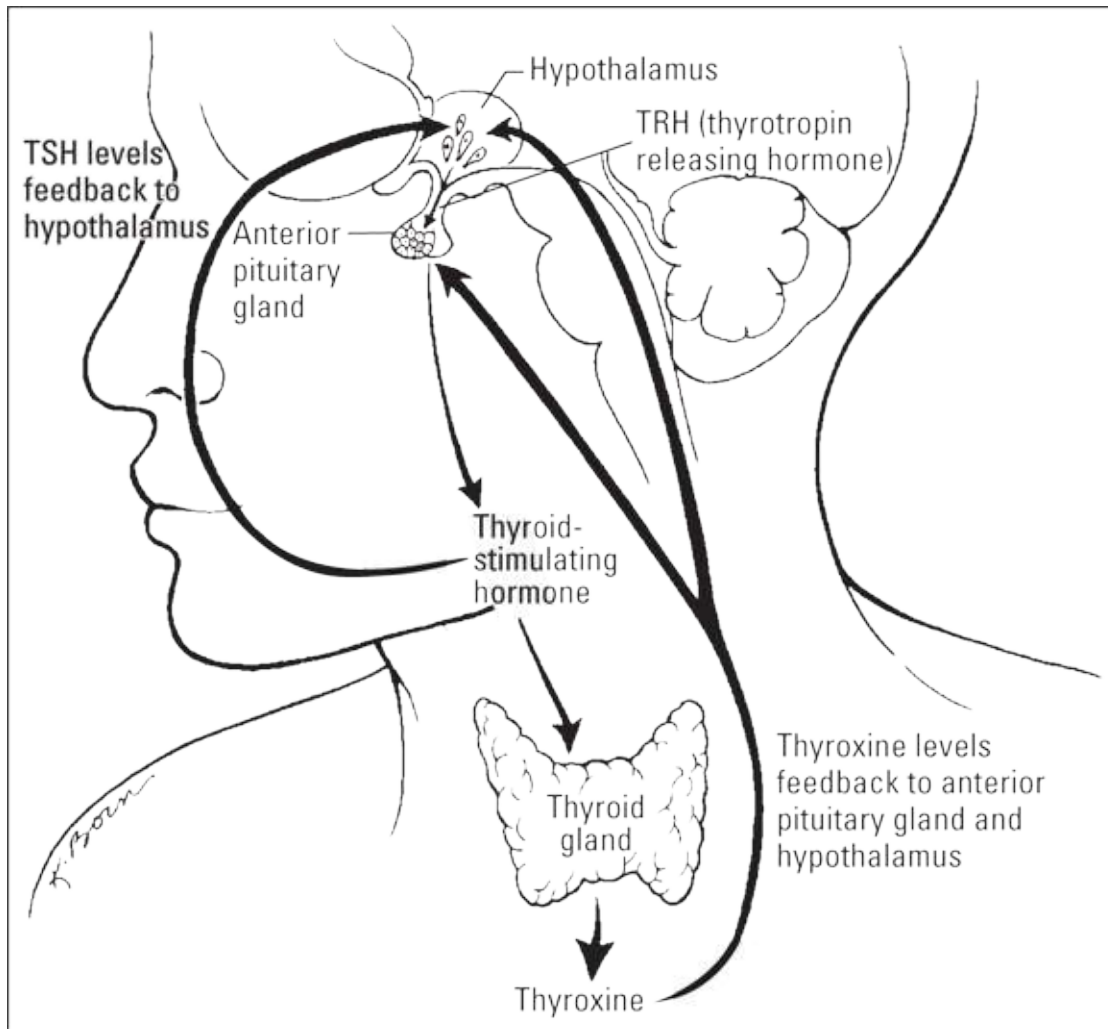


Illustration by Kathryn Born, MA

FIGURE 10-1: The working relationship of the hypothalamus and the pituitary gland.

The hypothalamus



REMEMBER The hypothalamus is the unsung hero linking the body's two primary control systems — the endocrine system and the nervous system. Part of the brain and part of the endocrine system, the hypothalamus is connected to the pituitary via a narrow stalk called the *infundibulum*. In its supervisory role, the hypothalamus provides neurohormones to control the pituitary gland's secretions. Releasing and inhibiting hormones produced by the hypothalamus regulate the release of most anterior pituitary hormones.

The hypothalamus sits just above the pituitary gland, which is nestled in the middle of the human head in a depression of the skull's sphenoid bone called the *sella turcica*.

The pituitary



REMEMBER The pituitary's *anterior lobe*, also called the *adenohypophysis*, is sometimes called the “master gland” because of its role in regulating and maintaining other endocrine glands. Hormones that act on other endocrine glands are called *tropic hormones*; all the hormones produced in the anterior lobe are polypeptides. Among the hormones produced in the anterior lobe of the pituitary gland are the following:

- » **Follicle-stimulating hormone (FSH):** Signals ovum (egg) development in the ovary and following ovulation, the production of estrogen. Negative feedback from the estrogen blocks further secretion of FSH. Guys, don't think you needn't worry about FSH: It's present in you, too, encouraging production and maturation of sperm. (For a review of the male and female reproductive systems, flip to [Chapters 16](#) and [17](#).)
- » **Luteinizing hormone (LH):** Stimulates the production of sex hormones in both males and females and triggers ovulation (release of the egg).
- » **Prolactin (PRL):** Promotes milk production in mammary glands; its role in males is still poorly understood.
- » **Thyrotropic hormone, or thyroid-stimulating hormone (TSH):** Controls the development and release of thyroid gland hormones *thyroxine* (T4) and *triiodothyronine* (T3). The hypothalamus regulates TSH secretion by secreting thyrotropin-releasing hormone (TRH).
- » **Adrenocorticotrophic hormone (ACTH), or corticotropin:** Regulates the development, maintenance, and secretions of the cortex of the adrenal gland. Its release is triggered by the hypothalamus in response to stress.
- » **Growth hormone (GH):** Stimulates body weight growth and regulates skeletal growth. This is the only hormone secreted by the anterior lobe that has a general effect on nearly every cell in the body. Its release is triggered by growth hormone releasing hormone (GHRH) from the

hypothalamus.



REMEMBER The *posterior lobe*, or *neurohypophysis*, of the pituitary gland stores and releases neurohormone secretions produced by the hypothalamus. This lobe is connected to the hypothalamus by the *hypophyseal tract*, axons with cell bodies lying in the hypothalamus. Whereas the adenohypophysis is made up of epithelial cells, the neurohypophysis is largely composed of nerve fibers and glial cells called *pituicytes*.

Among the hormones produced by the hypothalamus but released from the posterior lobe of the pituitary gland are the following:

- » **Oxytocin (OT):** Stimulates contraction of the uterus during childbirth and the release of breast milk in nursing women. Experts believe that intimate social activities such as hugging, touching, and sexual pleasures release oxytocin in both males and females leading to its role in bonding and its being dubbed “the love hormone.”
- » **Vasopressin, or antidiuretic hormone (ADH):** Constricts smooth muscle tissue in the blood vessels, elevating blood pressure and increasing the amount of water reabsorbed by the kidneys, which reduces the production of urine. The hypothalamus has specialized neurons called *osmoreceptors* that monitor the amount of solute in the blood.

See how much of this information you’re absorbing:



EXAMPLE Q. The hormone that stimulates ovulation is the

- a. follicle-stimulating hormone (FSH).
- b. antidiuretic hormone (ADH).
- c. oxytocin (OT).
- d. thyroid-stimulating hormone (TSH).
- e. luteinizing hormone (LH).

A. The correct answer is luteinizing hormone (LH). Don't be fooled into thinking it's FSH; that hormone does its job earlier, when it encourages a follicle to mature.

8-12 Mark the statement with a T if it's true or an F if it's false:

8. ____ The pituitary gland consists of two parts: an endocrine gland and modified nerve tissue.
9. ____ The pituitary gland is found in the sella turcica of the temporal bone.
10. ____ The adenohypophysis is called the master gland because of its influence on all the body's tissues.
11. ____ ADH causes constriction of smooth muscle tissue in the blood vessels, which elevates the blood pressure.
12. ____ The neurohypophysis stores and releases secretions produced by the hypothalamus.

13 Why is the hypothalamus considered to be just as important as the pituitary gland?

- a. It contracts and relaxes to regulate the so-called master gland.
- b. It brings in substances from the thyroid.
- c. It tells the pituitary gland what to release and when.
- d. It controls how large the other glands can grow.
- e. It influences the other glands more directly than the pituitary does.

14 Which of the following is not a pituitary hormone?

- a. Prolactin (PRL)
- b. Follicle-stimulating hormone (FSH)
- c. Growth hormone (GH)
- d. Progesterone (P4)
- e. Luteinizing hormone (LH)

The Supporting Cast of Glandular Characters

While the pituitary orchestrates the show at center stage, the endocrine system enjoys the support of a number of other important glands. Lying in various locations throughout the body, the glands in the following sections secrete check-and-balance hormones that keep the body in tune.

Topping off the kidneys: The adrenal glands

Also called *suprarenals*, the adrenal glands lie atop each kidney like little hats. The central area of each is called the *adrenal medulla*, and the outer layers are called the *adrenal cortex*. Each glandular area secretes different hormones. The cells of the cortex produce more than 30 steroids, including the hormones *aldosterone*, *cortisol*, and some sex hormones. The medullary cells secrete *epinephrine* (you may know it as *adrenaline*) and *norepinephrine* (also known as *noradrenaline*).

The cortex

Made up of closely packed epithelial cells, the adrenal cortex is loaded with blood vessels. Layers form an outer, middle, and inner zone of the cortex. Each zone is composed of a different cellular arrangement and secretes different steroid hormones.

The following are among the hormones produced by the cortex:

- » *Aldosterone* regulates electrolytes (sodium and potassium mineral salts) retained in the body. It promotes the conservation of water and reduces urine output in the kidneys.
- » *Cortisol* is released in response to stress helping to trigger the “fight-or-flight” response. It acts as an anti-inflammatory agent and can increase blood pressure. It also acts as an antagonist to insulin, increasing blood sugar.
- » *Androgens* and *estrogens* are cortical sex hormones. Androgens generally convey antifeminine effects, thus accelerating maleness, although in women adrenal androgens maintain the sexual drive. Too much androgen in females can cause *virilism* (male secondary sexual characteristics).

Estrogens have the opposite effect, accelerating femaleness. Too much estrogen in a male produces feminine characteristics.

The medulla

The adrenal medulla is made of irregularly shaped *chromaffin cells* arranged in groups around blood vessels. The sympathetic division of the autonomic nervous system controls these cells as they secrete adrenaline and noradrenaline. The adrenal medulla produces approximately 80 percent adrenaline and 20 percent noradrenaline. Both hormones have similar molecular structure and physiological functions:

- » **Adrenaline** accelerates the heartbeat, stimulates respiration, slows digestion, increases muscle efficiency, and helps muscles resist fatigue.
- » **Noradrenaline** does similar things but also raises blood pressure by stimulating constriction of muscular arteries.



TIP

The terms “adrenaline” and “noradrenaline” are interchangeable with the terms “epinephrine” and “norepinephrine.” You’re likely to encounter both in textbooks and exams.

Thriving with the thyroid

The largest of the endocrine glands, the thyroid is shaped like a large butterfly with two lobes connected by a fleshy *isthmus* positioned in the front of the neck, just below the larynx and on either side of the trachea (you can see the thyroid in [Figure 10-1](#)). In response to TSH from the anterior pituitary, the thyroid releases its two primary hormones: *thyroxine* and *triiodothyronine*. These regulate the body’s metabolic rate by acting on nearly every cell in the body.



WARNING

The thyroid produces both T3 and T4, but only T3 is the active form of the hormone. Because they are lipid-soluble, they enter the cell where T4 can be converted to T3 by enzymes. So why does the thyroid even both making T4? It has a longer half-life, meaning it will circulate in the

bloodstream for longer than T3.

Parafollicular cells (also called *C cells*) in the thyroid secrete another hormone: *calcitonin*. This hormone promotes bone growth by stimulating osteoblasts (bone-building cells) to take up calcium from the bloodstream and deposit it into the matrix. High blood calcium levels stimulate the secretion of more calcitonin.

Pairing up with the parathyroid

The parathyroid consists of four pea-sized glands that are attached to the posterior side of the thyroid gland. These secrete *parathyroid hormone* (PTH), which does the opposite of calcitonin. When blood calcium levels dip, the parathyroid secretes PTH, which increases calcium absorption from the intestine, and triggers osteoclasts to release calcium stored in the bone (by breaking down the matrix). The homeostasis of blood calcium is critical to the conduction of nerve impulses, muscle contraction, and blood clotting.

Pinging the pineal gland

The pineal gland, also called the *epiphysis cerebri*, is a small, oval gland thought to play a role in regulating the body's biological clock. It lies between the cerebral hemispheres and is attached to the thalamus near the roof of the third ventricle.



REMEMBER Because it both secretes a hormone and receives visual nerve stimuli, the pineal gland is considered part of both the nervous system and the endocrine system. Its hormone *melatonin* is believed to play a role in circadian rhythms, the pattern of repeated behavior associated with the cycles of night and day. The pineal gland is affected by changes in light; when its photoreceptors begin being stimulated less and less, it secretes melatonin, making you feel sleepy.

Thumping the thymus

The thymus secretes a peptide hormone called *thymosin*, which promotes the production and maturation of T lymphocyte cells. T cells are a type of white blood cell that play a key role in immunity — mounting a targeted attack against the invading germ. This is discussed further in [Chapter 12](#). The gland

is large in children and atrophies with age because the “training” of T cells is mostly complete by the onset of puberty.

Pressing the pancreas

The pancreas is both an exocrine and an endocrine gland, which means that it secretes some substances through ducts while others go directly into the bloodstream. (We cover its exocrine functions in [Chapter 14](#).) The pancreatic endocrine glands are clusters of cells called the *islets of Langerhans*. Within the islets are a variety of cells, including

- » **A cells (alpha cells)** that secrete the hormone *glucagon* to increase blood sugar by triggering *glycogen* breakdown in the liver (glycogen is the storage form of glucose).
- » **B cells (beta cells)** that secrete *insulin* to decrease blood sugar levels by triggering the liver to store glucose, increases lipid synthesis, and stimulates protein synthesis.
- » **D cells (delta cells)** that secrete *somatostatin*, to inhibit the secretion of insulin and glucagon.
- » **F cells (PP cells)** that secrete a pancreatic polypeptide that regulates the release of pancreatic digestive enzymes.



WARNING

We tend to think of insulin as a necessity to keep blood sugar level down in the normal range, which it does. But its mechanism in doing so is more important. When insulin binds to receptors on a cell’s surface, it triggers the opening of glucose channels, allowing glucose to diffuse into the cell. Without insulin, cells cannot obtain enough glucose for cellular respiration and thus cannot produce enough ATP to carry out their functions.

See whether all this information has your hormones raging:

15-19 Mark the statement with a T if it’s true or an F if it’s false:

15. _____ The adrenal glands are located in the cortex of the kidneys.
16. _____ Adrenaline is functional in the absorption of stored carbohydrates

and fat.

17. _____ Aldosterone is functional in regulating the amount of insulin in the body.
18. _____ The sympathetic division of the autonomic nervous system controls the cells of the adrenal medulla.
19. _____ The adrenal cortex produces sex hormones.

20 The thymus produces thymosin, which

- a. stimulates the pineal gland.
- b. inhibits the pituitary gland.
- c. initiates lymphocyte development.
- d. stimulates the thyroid gland.
- e. controls the hypothalamus.

21-24 Mark the statement with a T if it's true or an F if it's false.

21. _____ Parafollicular cells of the thyroid produce hormones that affect the metabolic rate of the body.
22. _____ Thyroxine (T4) is normally secreted in lower quantity than triiodothyronine (T3).
23. _____ The hormone calcitonin helps regulate the concentration of sodium and potassium.
24. _____ The parathyroid gland contains cells that secrete parathyroid hormone (PTH), which works to increase blood calcium levels.

25 Which statement is NOT true of the pineal gland?

- a. It secretes melatonin.
- b. Nerve fibers stimulate the pineal cells.
- c. As light decreases, secretion increases.
- d. It promotes immunity.
- e. It's considered part of both the endocrine and nervous systems.

26 The adrenal medulla serves one primary function. What is it?

- a. To produce erythrocyte cells
- b. To absorb excess iodine from the bloodstream
- c. To produce epinephrine
- d. To secrete polypeptides
- e. To regulate water retention in the kidneys

27 Clusters of cells can be found within the islets of Langerhans in the pancreas. Which variety of cell secretes glucagon to increase blood sugar?

- a. Beta cells, also known as B cells
- b. Alpha cells, also known as A cells
- c. PP cells, also known as F cells
- d. Delta cells, also known as D cells

28-37 Use the terms that follow to identify the structures of the endocrine system shown in [Figure 10-2](#).

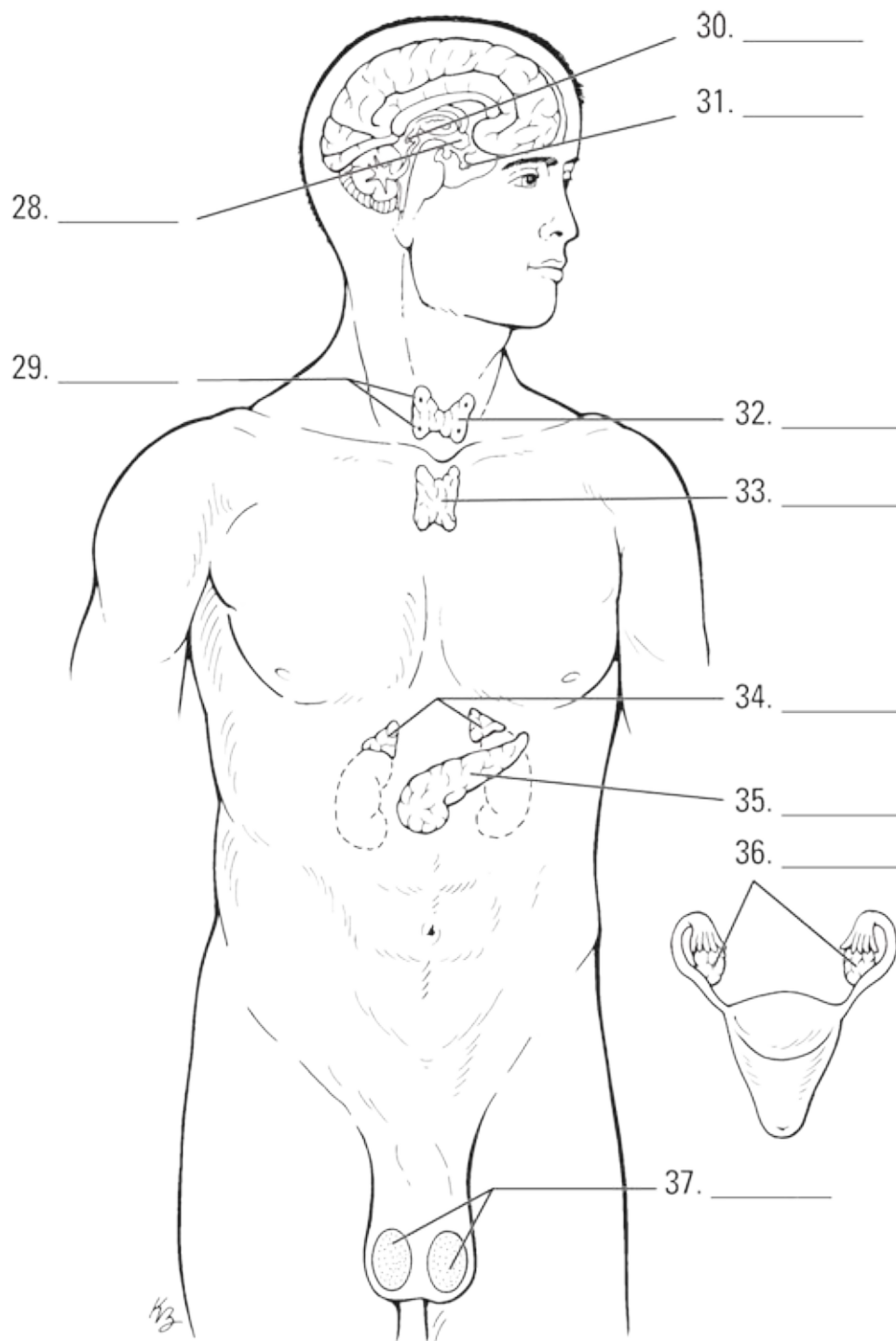


Illustration by Kathryn Born, MA

FIGURE 10-2: The endocrine system.

- a. Thyroid gland
- b. Pineal gland
- c. Pituitary gland

- d. Adrenal glands
- e. Ovaries
- f. Parathyroid gland
- g. Testes
- h. Hypothalamus
- i. Pancreas
- j. Thymus

Dealing with Stress: Homeostasis

Nothing upsets your delicate cells more than a change in their internal environment. A stimulus such as fear or pain provokes a response that upsets your body's carefully maintained equilibrium. Such a change initiates a nerve impulse to the hypothalamus that activates the sympathetic division of the autonomic nervous system and increases secretions from the adrenal glands. This change — called a *stressor* — produces a condition many know all too well: *stress*. Stress is anything that disrupts the normal function of the body. The body's immediate response is to push for *homeostasis* — keeping everything the same inside. Inability to maintain homeostasis — a condition known as *homeostatic imbalance* — may lead to disease or even death.

The body's effort to maintain homeostasis invokes a series of reactions called the *general adaptation syndrome*, the predictable phases of response to stress that's initiated by the hypothalamus. When the hypothalamus receives stress information, it responds by preparing the body for fight or flight — in other words, some kind of decisive, immediate, physical action. This reaction increases blood levels of glucose, glycerol, and fatty acids; increases the heart rate and breathing rate; redirects blood from skin and internal organs to the skeletal muscles; and increases the secretion of adrenaline from the adrenal medulla. The hypothalamus releases *corticotropin-releasing hormone* (CRH) that stimulates the anterior lobe of the pituitary to secrete *adrenocorticotrophic hormone* (ACTH), which tells the adrenal cortex to secrete more cortisol. That cortisol supplies the body with amino acids and an extra energy source needed to repair any injured tissues that may result from the impending crisis.

As part of the general stress syndrome, the pancreas produces glucagon, and

the anterior pituitary secretes growth hormones, both of which prepare energy sources and stimulate the absorption of amino acids to repair damaged tissue. The posterior pituitary secretes antidiuretic hormone, making the body hang on to sodium ions and spare water. The subsequent decrease in urine output is important to increase blood volume, especially if there is bleeding or excessive sweating.

Wow. With the body gearing up like that every time, it's no wonder that people subjected to repeated stress are often sickly.

We'll try not to stress you out with these practice questions:

38-41 Mark the statement with a T if it's true or an F if it's false.

- 38. ____ The hypothalamus initiates the general adaptation syndrome.
- 39. ____ During stress, the pancreas produces thyroxine (T4).
- 40. ____ Homeostatic imbalance has primarily positive effects on the body.
- 41. ____ Cortisol released in reaction to stress provides an extra energy source for tissue repair.

42 When changes occur in the body's internal environment, a hormonal chain reaction occurs, beginning with the _____, which prompts secretions from the _____, which in turn instruct the _____ to secrete cortisol.

- a. hypothalamus; adrenal medulla; pituitary
- b. pituitary; hypothalamus; adrenal medulla
- c. hypothalamus; pituitary; adrenal cortex
- d. pituitary, hypothalamus, adrenal cortex
- e. adrenal medulla; hypothalamus; pituitary

43 The body's initial reaction to a stressor is

- a. the fight-or-flight response.
- b. the repair response.
- c. to promote rapid wound healing.

- d. the stress reflex.
- e. to promote normal metabolism.

Answers to Questions on the Endocrine System

The following are answers to the practice questions presented in this chapter.

- ① The endocrine system brings about changes in the metabolic activities of the body tissue. **True.** Metabolism is one of the areas influenced by hormones.
- ② The amount of hormone released is determined by the body's need for that hormone at the time. **True.** In many ways, it's a self-regulating system: Just enough hormones are distributed to balance everything else out.
- ③ The glands of the endocrine system are composed of cartilage cells. **False.** That connection makes no sense whatsoever.
- ④ Endocrine glands aren't functional in reproductive processes. **False.** The endocrine system is a key component in reproduction.
- ⑤ Some hormones can be derivatives of amino acids, whereas others are synthesized from cholesterol. **True.** Amino acids work for nonsteroid hormones, and cholesterol does the trick for steroid-based hormones.
- ⑥ **a. Exocrine** glands secrete their product through ducts while **endocrine** glands secrete their product into the interstitial fluid, which flows into the blood.
- ⑦ **Steroid hormones are derived from cholesterol making them lipid-soluble. As a result, steroid hormones can easily enter a cell, find their way to the nucleus, and directly stimulate (or inhibit) protein production. Nonsteroid hormones are water-soluble, so they don't have it so easy — a cell would have to invest a great deal of energy to bring it in. Instead, they bind to receptors, which triggers a chain reaction of events to elicit the target cell's response.**

- 8) The pituitary gland consists of two parts: an endocrine gland and modified nerve tissue. **True.** Remember that the anterior lobe is mostly epithelial cells, whereas the posterior lobe contains primarily nerves (bundled axons).
- 9) The pituitary gland is found in the sella turcica of the temporal bone. **False.** The pituitary gland is found in the sella turcica but that's part of the sphenoid bone, not the temporal bone.
- 10) The adenohypophysis is called the master gland because of its influence on all the body's tissues. **False.** It earned the title "master gland" because of its influence over the other endocrine glands.
- 11) ADH causes constriction of smooth muscle tissue in the blood vessels, which elevates the blood pressure. **True.** It's understandable that constriction increases pressure. This, however, is a secondary function to its influence on the kidneys to retain water.
- 12) The neurohypophysis stores and releases secretions produced by the hypothalamus. **True.** Seems a strange thing for a structure made of nerves to do, but it does its job well.
- 13) Why is the hypothalamus considered to be just as important as the pituitary gland? **c. It tells the pituitary gland what to release and when.** If you remember that the pituitary gland is attached to the bottom of the hypothalamus, you can see the importance of the relationship between the two.
- 14) Which of the following is not a pituitary hormone? **d. Progesterone.** That's made by the corpus luteum in the ovary following ovulation.
- 15) The adrenal glands are located in the cortex of the kidneys. **False.** They're atop the kidneys.
- 16) Adrenaline is functional in the absorption of stored carbohydrates and fat. **False.** Adrenaline does lots of things, but not that.
- 17) Aldosterone is functional in regulating the amount of insulin in the body. **False.** This hormone regulates the uptake of mineral salts and thus the conservation of water by the kidneys.

- 18) The sympathetic division of the autonomic nervous system controls the cells of the adrenal medulla. **True.** Those cells are busy secreting both adrenaline and noradrenaline.
- 19) The adrenal cortex produces sex hormones. **True.** Production of these hormones is not limited to the gonads.
- 20) The thymus produces thymosin, which **c. initiates lymphocyte development.** Specifically, the production and training of T cells.
- 21) Parafollicular cells of the thyroid produce hormones that affect the metabolic rate of the body. **False.** These are the cells that produce calcitonin.
- 22) Thyroxine (T4) is normally secreted in a lower quantity than triiodothyronine (T3). **False.** In fact, it's just the opposite — more T4 is secreted than T3 because it lasts longer.
- 23) The hormone calcitonin helps regulate the concentration of sodium and potassium. **False.** Actually, calcitonin lowers plasma calcium and phosphate levels.
- 24) The parathyroid gland contains cells that secrete parathyroid hormone (PTH) which works to increase blood calcium levels. **True.** As far as we know, PTH is the only hormone that can trigger the breakdown of bone matrix to release the calcium stored there.
- 25) Which statement is NOT true of the pineal gland? **d. It promotes immunity.** It's more of a biological-clock kind of gland.
- 26) The adrenal medulla serves one primary function. What is it? **c. To produce epinephrine.** That's about all the adrenal medulla does. (Actually, it produces 80 percent epinephrine and 20 percent norepinephrine.)
- 27) Clusters of cells can be found within the islets of Langerhans in the pancreas. Which variety of cell secretes glucagon to increase blood sugar? **b. Alpha cells, also known as A cells.** The secretion from A cells increases blood sugar, insulin from B cells decreases blood sugar, somatostatin from D cells inhibits both insulin and glucagon, and the polypeptide from F cells

regulates digestive enzymes.

28-37) Following is how [Figure 10-2](#), the endocrine system, should be labeled.

28. **h. Hypothalamus**; 29. **f. Parathyroid gland**; 30. **b. Pineal gland**; 31. **c. Pituitary gland**; 32. **a. Thyroid gland**; 33. **j. Thymus**; 34. **d. Adrenal glands**; 35. **i. Pancreas**; 36. **e. Ovaries**; 37. **g. Testes**.

38) The hypothalamus initiates the general adaptation syndrome. **True**. It does so by initiating the body's fight or flight response.

39) During stress, the pancreas produces thyroxine (T4). **False**. That's the thyroid's job, and T4 has nothing to do with stress, anyway.

40) Homeostatic imbalance has primarily positive effects on the body. **False**. Imbalance is a bad thing — so bad that it's capable of killing.

41) Cortisol released in reaction to stress provides an extra energy source for tissue repair. **True**. It's part of how the body gears up to manage an impending crisis.

42) When changes occur in the body's internal environment, a hormonal chain reaction occurs, beginning with the **c. hypothalamus**, which prompts secretions from the **pituitary**, which instruct the **adrenal cortex** to secrete cortisol.

43) The body's initial reaction to a stressor is **a. the fight-or-flight response**. You're either going to put up your dukes or run like the wind — or freeze, as some psychologists argue that it should be renamed the fight-flight-or-freeze response.

Part 4

Feed and Fuel: Supply and Transport

IN THIS PART ...

Take the pulse of the cardiovascular system and discover how its internal transit routes carry nutrients and oxygen to every nook and cranny of the body.

Delve into the lymphatic ducts and find out about the body's internal defense forces.

Breathe in the details of the respiratory system, including the parts of the respiratory tract (the nose, the sinuses, the throat, and the lungs) and how oxygen finds its way into the bloodstream.

Fuel up with food and then study what happens to it in the digestive system after every possible nutrient has been wrung from it.

Bat cleanup with the kidneys and track how they remove waste from the body as part of the urinary system.

Chapter 11

Spreading the Love: The Cardiovascular System

IN THIS CHAPTER

- » Understanding the heart's rhythm and structure
 - » Identifying the heart's chambers
 - » Checking out the heart's conduction system
 - » Tracing arteries, veins, and capillaries
-

This chapter gets to the heart of the well-oiled human machine to see how its central pump is the hardest-working muscle in the entire body. From a month after you're conceived to the moment of your death, this phenomenal powerhouse pushes a liquid connective tissue — blood — and its precious cargo of oxygen and nutrients to every nook and cranny of the body, and then it keeps things moving to bring carbon dioxide and waste products back out again. In the first seven decades of human life, the heart beats roughly 2.5 billion times. Do the math: How many pulses has your ticker clocked if the average heart keeps up a pace of 72 beats per minute, 100,000 per day, or roughly 36 million per year?

Moving to the Beat of a Pump

The cardiovascular system includes the heart, all the blood vessels, and the blood that flows through them. While it is still sometimes referred to as the *circulatory system*, that terminology is falling out of favor as our body circulates another fluid, called lymph, through a different system. ([Chapter 12](#) discusses the lymphatic system.) The cardiovascular system functions as a closed double loop. Closed because the blood never leaves the vessels; double because there are two distinct circuits (see [Figure 11-1](#)).